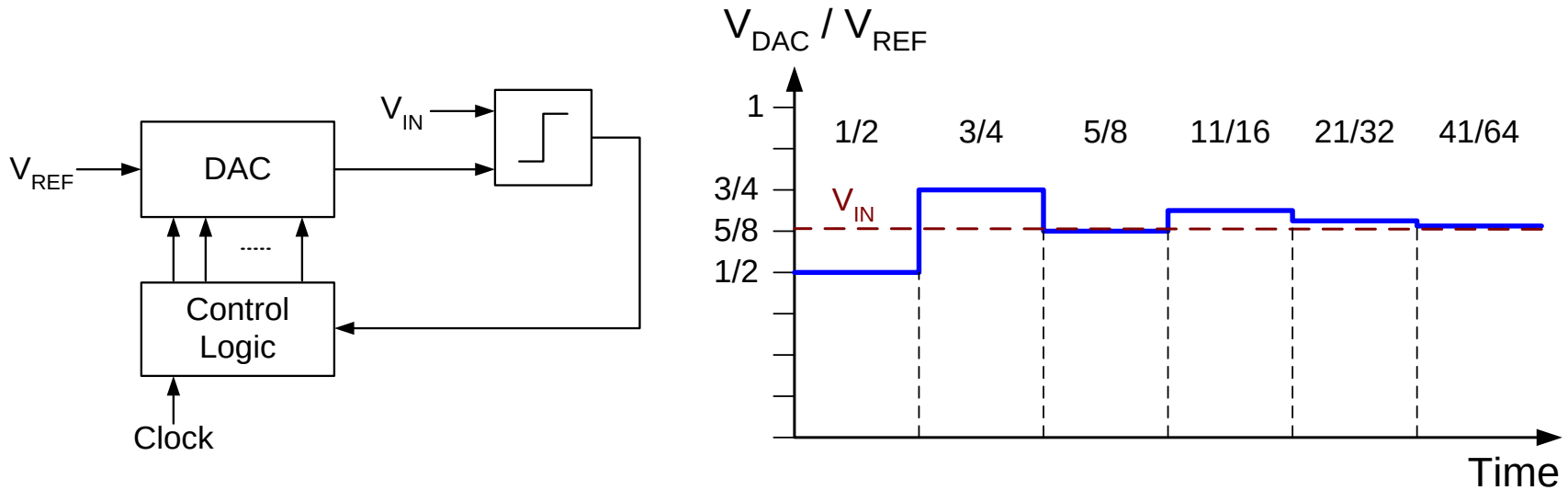


Successive Approximation Register (SAR) ADC

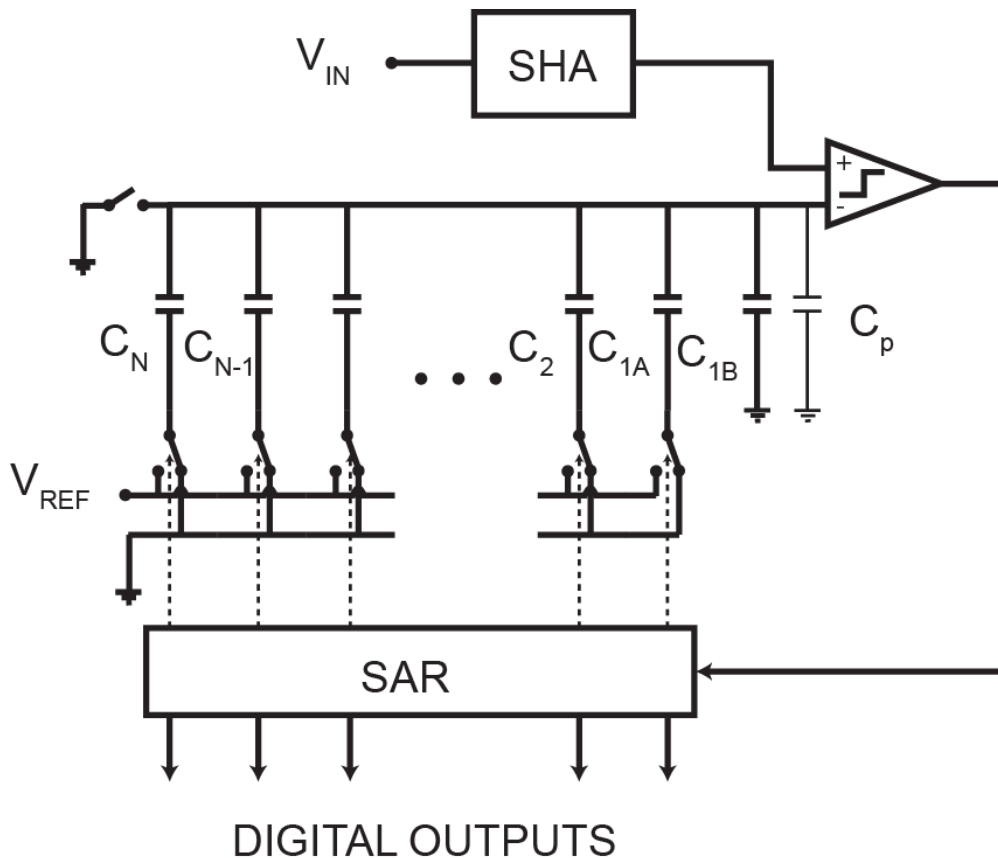
Nan Sun
Tsinghua University
nansun@Tsinghua.edu.cn

SAR ADC



- Binary search over DAC output
- High accuracy achievable
 - Rely on highly accurate comparator
- Moderate speed

Capacitive DAC Direct Implementation



$$C_{1A} = C_{1B} = C_u$$

$$C_2 = 2^1 C_u$$

$$C_3 = 2^2 C_u$$

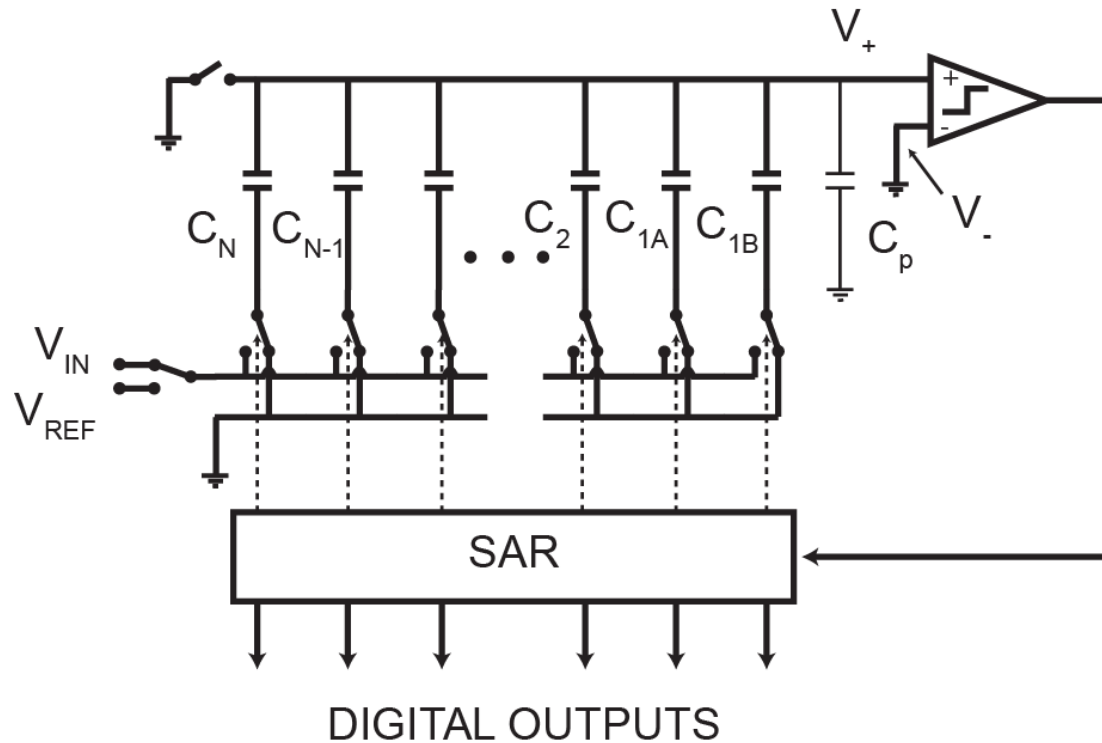
.

$$C_N = 2^{N-1} C_u$$

- Binary-weighted capacitor array DAC provides better matching than R-2R Ladder
- Nonlinear parasitic capacitance C_p is a problem: gain error and INL

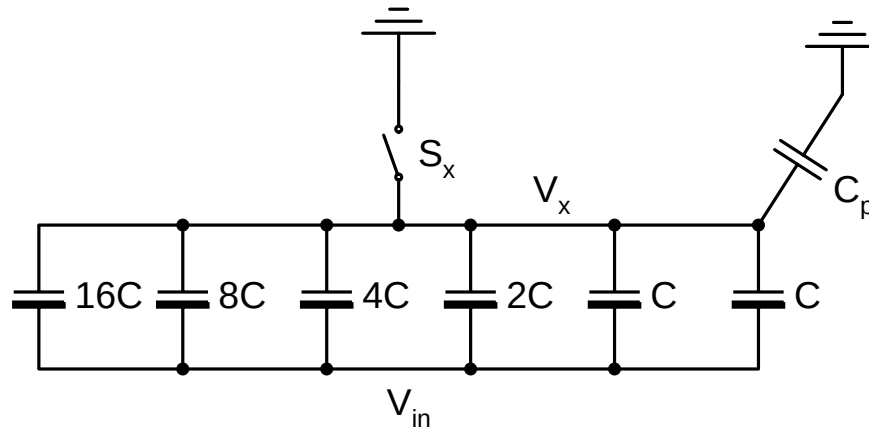
Implementation

- See e.g. [McCreary, JSSC 12/1975]



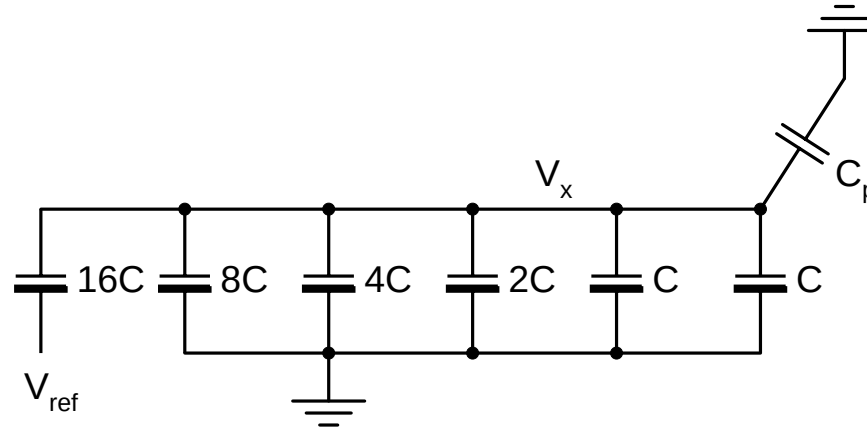
- Inherent S/H function (saves power/area/noise)
- Comparator common-mode voltage at GND
- Nonlinear parasitic capacitance C_p has no effect

Sampling Phase (5-bit Example)



- Total charge at node V_x after opening S_x

$$Q = -V_{in} \cdot 32C = -V_{in} \cdot C_{total}$$

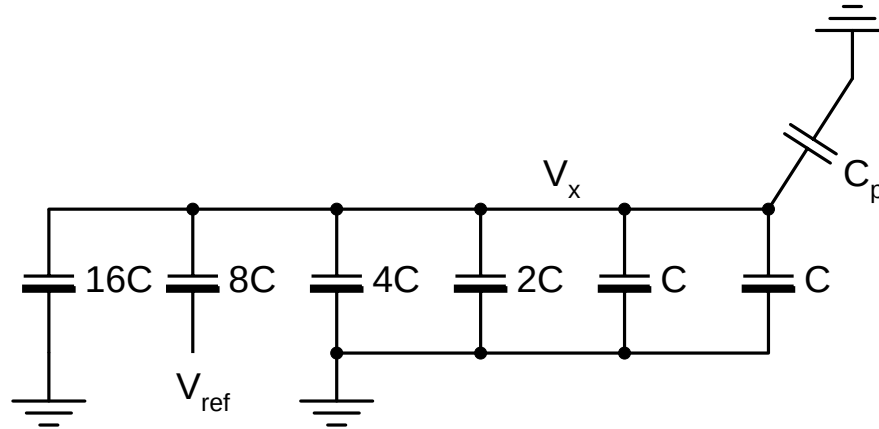


$$Q = -V_{in} \cdot C_{total} = (V_x - V_{ref}) \cdot 16C + V_x \cdot (16C + C_p)$$

$$\therefore V_x = \left(\frac{1}{2} V_{ref} - V_{in} \right) \cdot \frac{C_{total}}{C_{total} + C_p}$$

- $V_x < 0 \Rightarrow V_{in} > 0.5V_{ref} \Rightarrow \text{Bit5}=1$
- $V_x > 0 \Rightarrow V_{in} < 0.5V_{ref} \Rightarrow \text{Bit5}=0$

Bit4 Test (Assuming Bit5=0)



$$Q = -V_{in} \cdot C_{total} = (V_x - V_{ref}) \cdot 8C + V_x \cdot (24C + C_p)$$

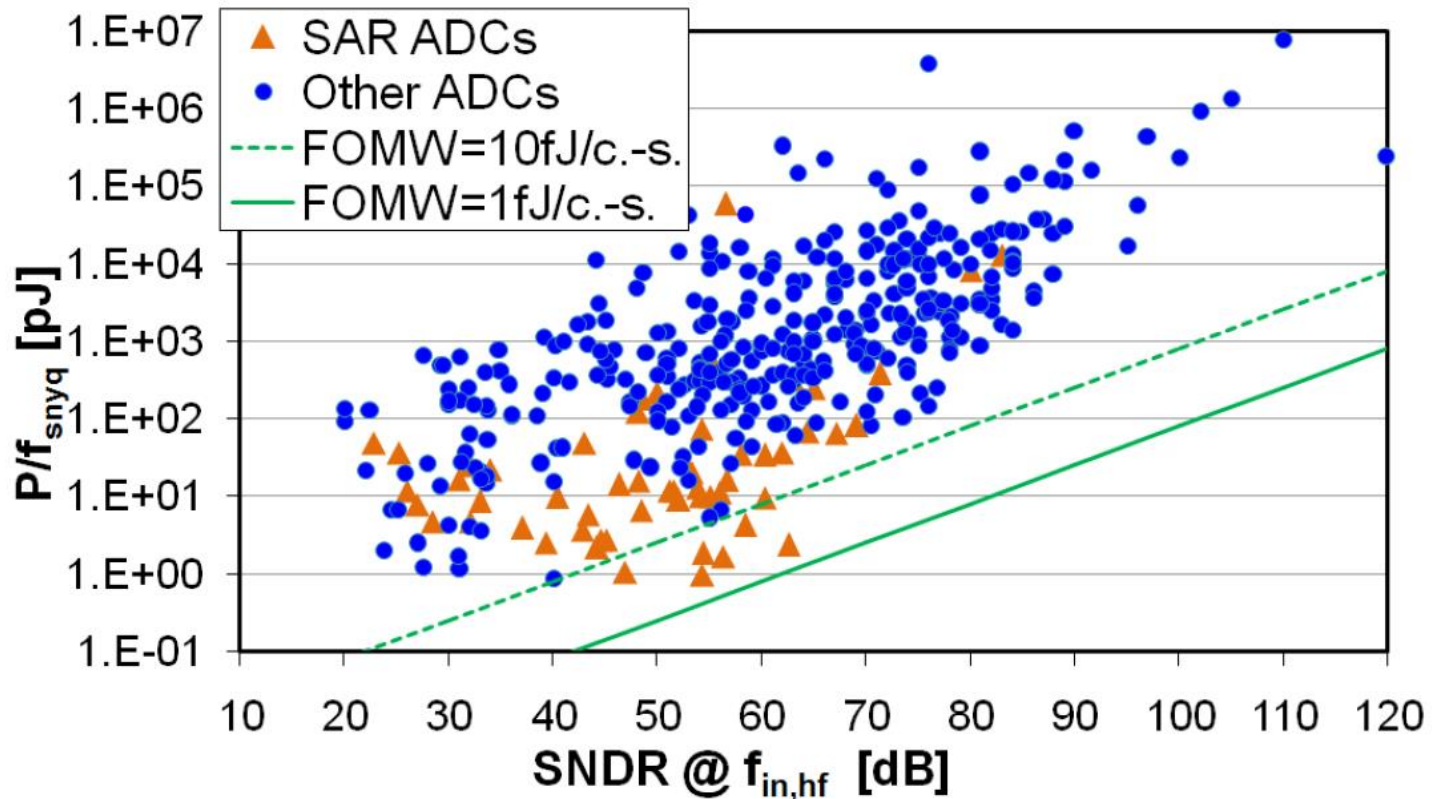
$$\therefore V_x = \left(\frac{1}{4} V_{ref} - V_{in} \right) \cdot \frac{C_{total}}{C_{total} + C_p}$$

- $V_x < 0 \Rightarrow V_{in} > 0.25V_{ref} \Rightarrow \text{Bit4}=1$
- $V_x > 0 \Rightarrow V_{in} < 0.25V_{ref} \Rightarrow \text{Bit4}=0$

Advantages

- SAR ADC is very simple!
 - A capacitive DAC
 - A comparator
 - A SAR logic
- SAR ADC is very power efficient!
 - Digital friendly
 - No static power

SAR ADC reaching <1fJ/conv-step



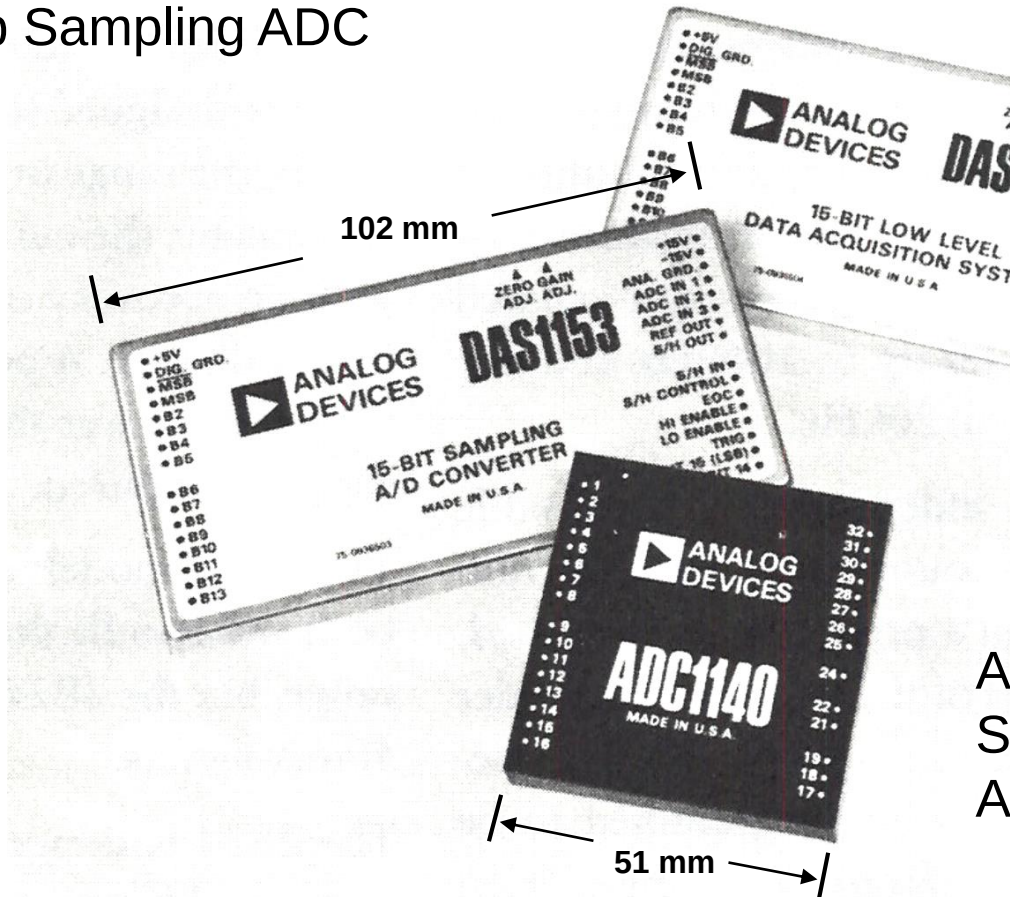
B. Murmann, "ADC Performance Survey 1997-2013. "
<http://www.stanford.edu/~murmann/adcsurvey.html>.

Limitations

- Conversion rate is lower than pipelined or flash ADC
 - N conversion cycles
 - DAC RC settling, comparator delay, SAR logic delay
- Capacitive DAC can be quite big for high resolution. If unit capacitor size is fixed, we can reduce the C_{total} by using:
 - “Two-stage” capacitor network [Yee, JSSC 8/79]
 - Multiple reference voltages generated by a resistor ladder
- Thermal noise set a limit for C_{total}
 - E.g. 16-bit resolution, $C_{\text{total}} \sim 100\text{pF}$ for kT/C noise
- If matching is an issue, an even larger C_{total} may be needed
 - E.g. if matching dictates $C_{\text{min}} = 10\text{fF}$, then $2^{16}C_{\text{min}} = 655\text{pF}$
 - Better solution is to use calibration [Lee, JSSC 12/84]

State of The Art ADCs circa 1982

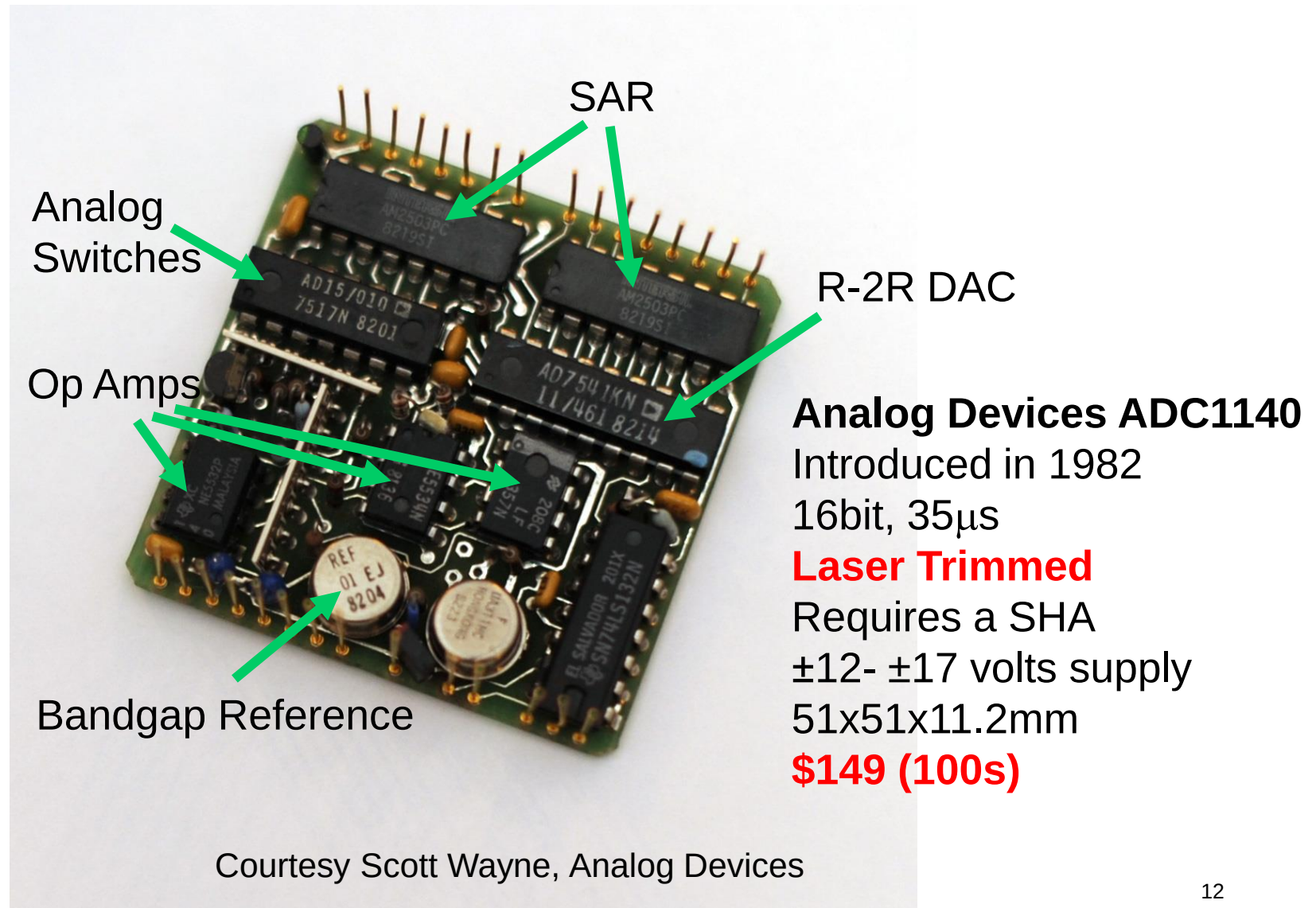
DAS1153
15b Sampling ADC



ADC1140 16b 30kS/s
Successive
Approximation ADC

Source: Analog Dialog vol. 16, Nov. 1982, Analog Devices, MA
Courtesy Scott Wayne, Analog Devices

16b ADC, Inside

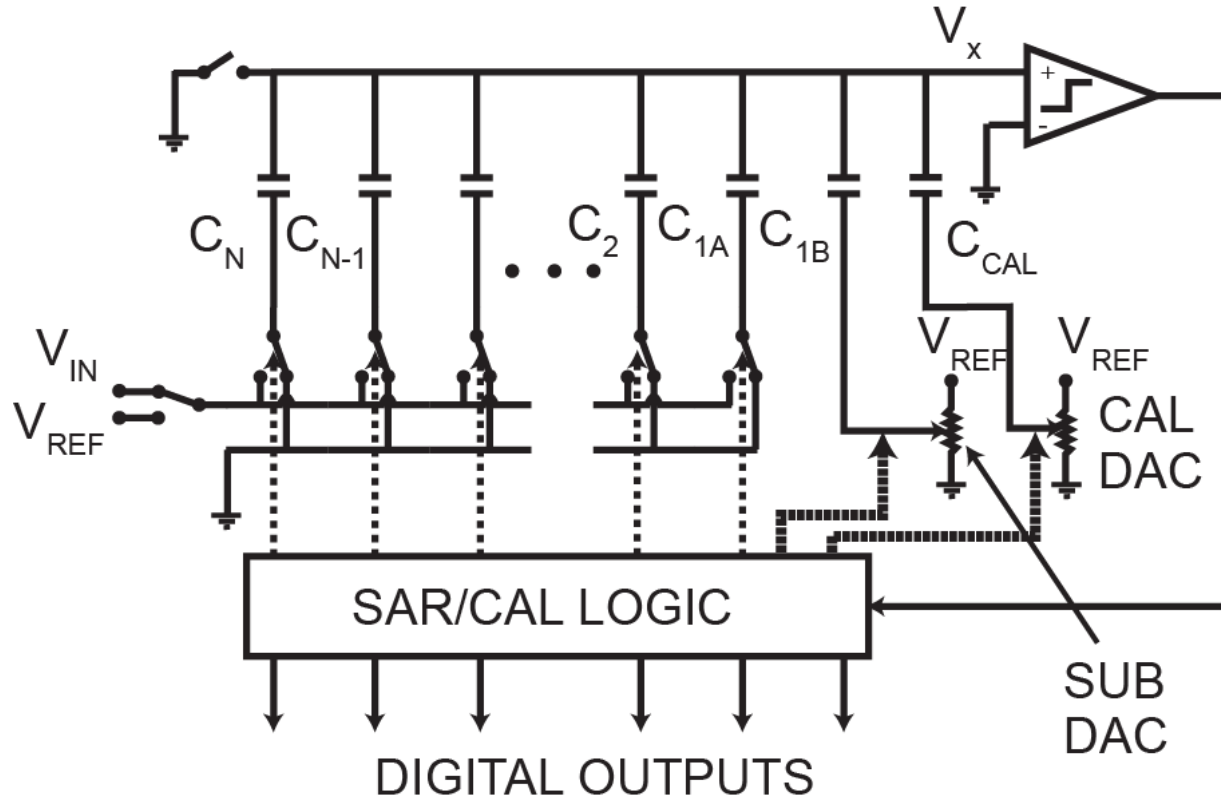


Charge-Redistribution ADC (1975)

Technology	10 μ NMOS
Resolution	10 bits
Linearity	$\pm 0.5 \text{ LSB}_{10}$
Conversion Rate	44 kS/s
Input Range	0-10 V
Array Capacitance	240 pF
Die Area	5.2 mm²

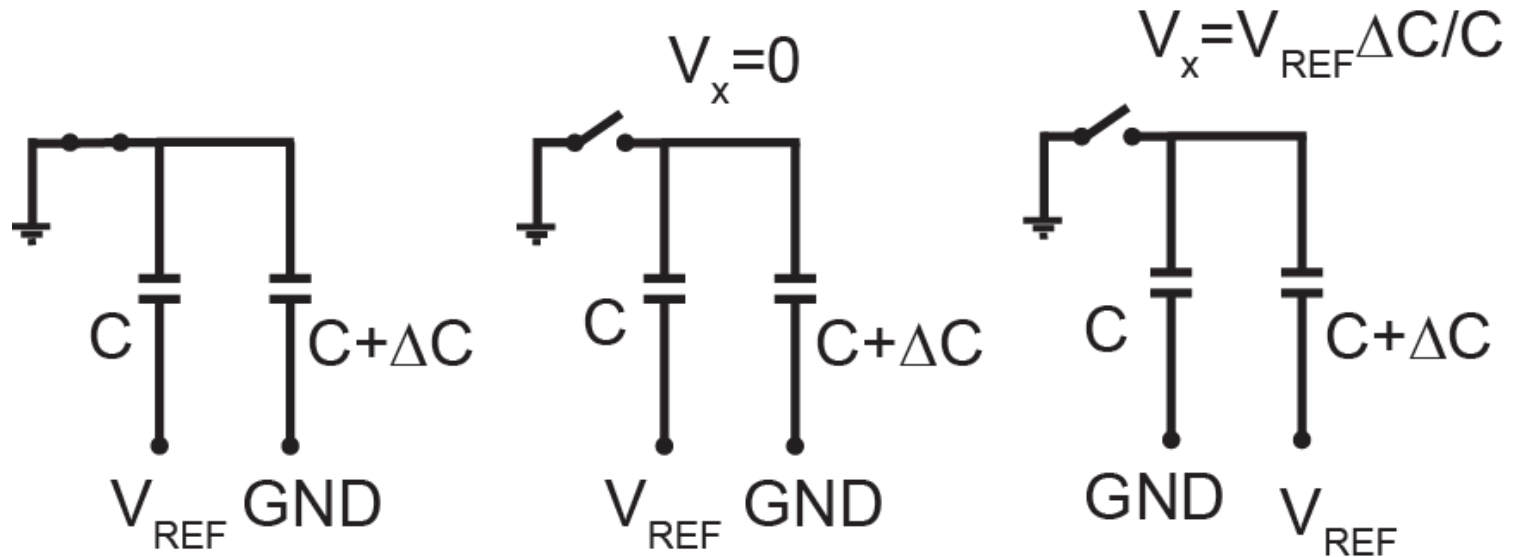
Reference: J. L. McCreary and P. R. Gray, "All MOS charge-redistribution analog-to-digital conversion techniques— Part I," IEEE J. Solid-State Circuits vol. SC-10, pp. 371-379, Dec. 1975.

Self-Calibrating ADC Block Diagram



- 10 bit plus sign capacitor array MAIN DAC
- 5 bit resistor string SUB DAC
- 7 bit resistor string calibration (CAL) DAC
- Accuracy is determined by MAIN DAC
- Digital calibration applied to MAIN DAC

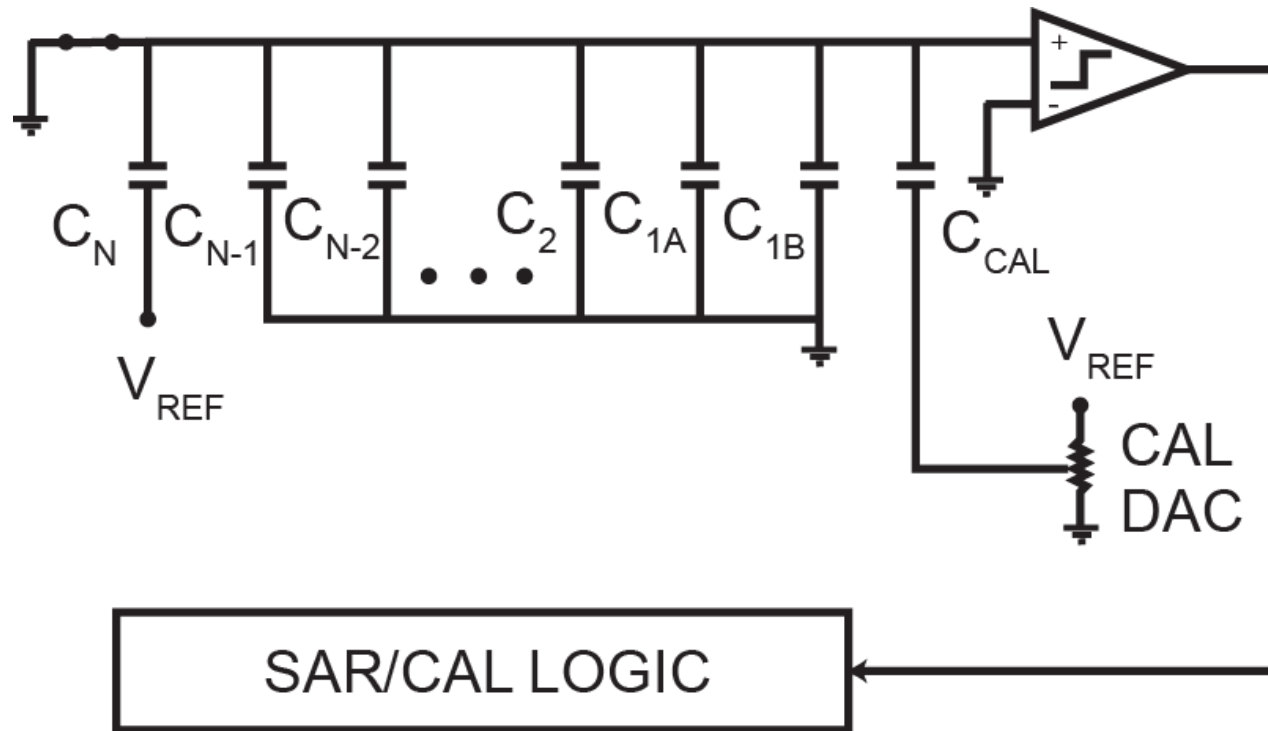
Basic Principle of Mismatch Measurements



After swapping the bottom plate voltages, the top plate voltage is proportional to the ratio match

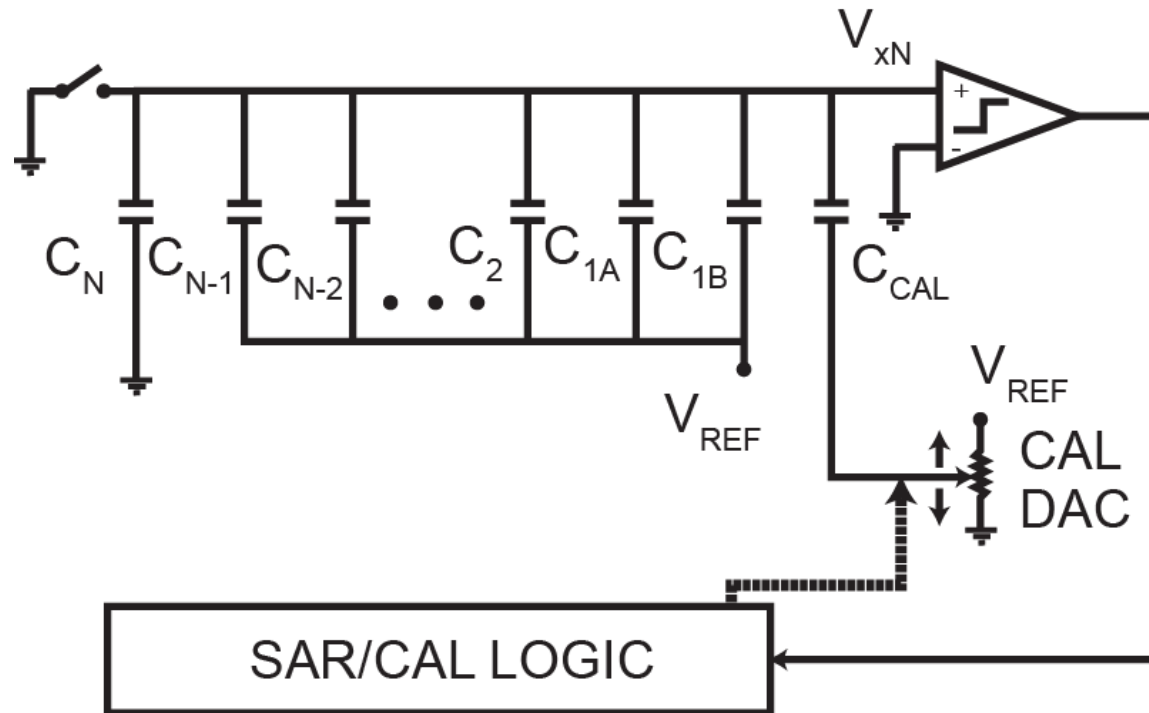
Reference: J. L. McCreary and D.A. Sealer, "Precision Capacitor Ratio Measurement Technique for Integrated Circuit Capacitor Arrays," IEEE Trans. Instrum. Meas. vol. IM-28, Mar. 1979.

Calibration: Precharge Phase (MSB)



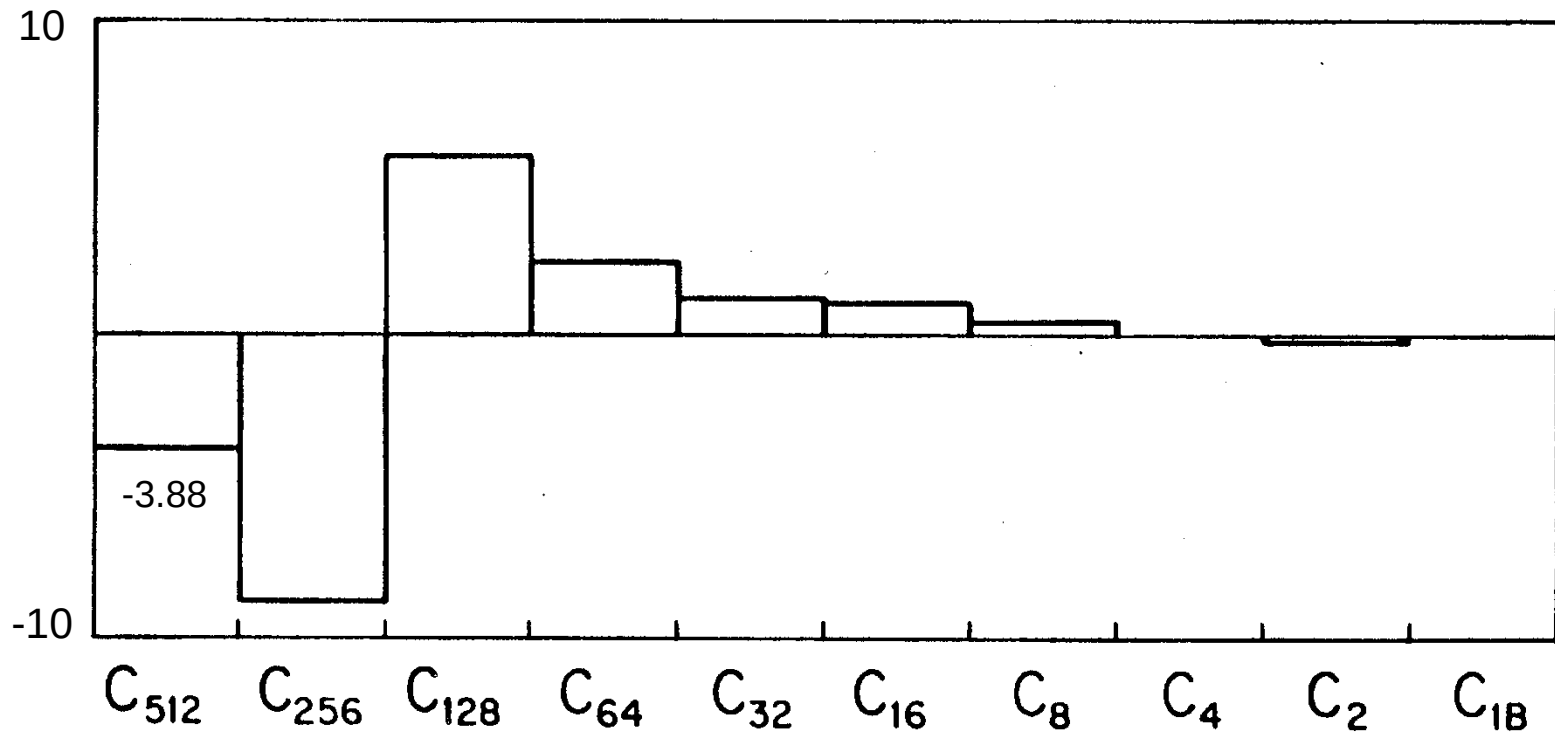
- MSB cap is charged to V_{REF} , the rest to GND
- CAL DAC is set to center tap

Mismatch Measurement Phase (MSB)



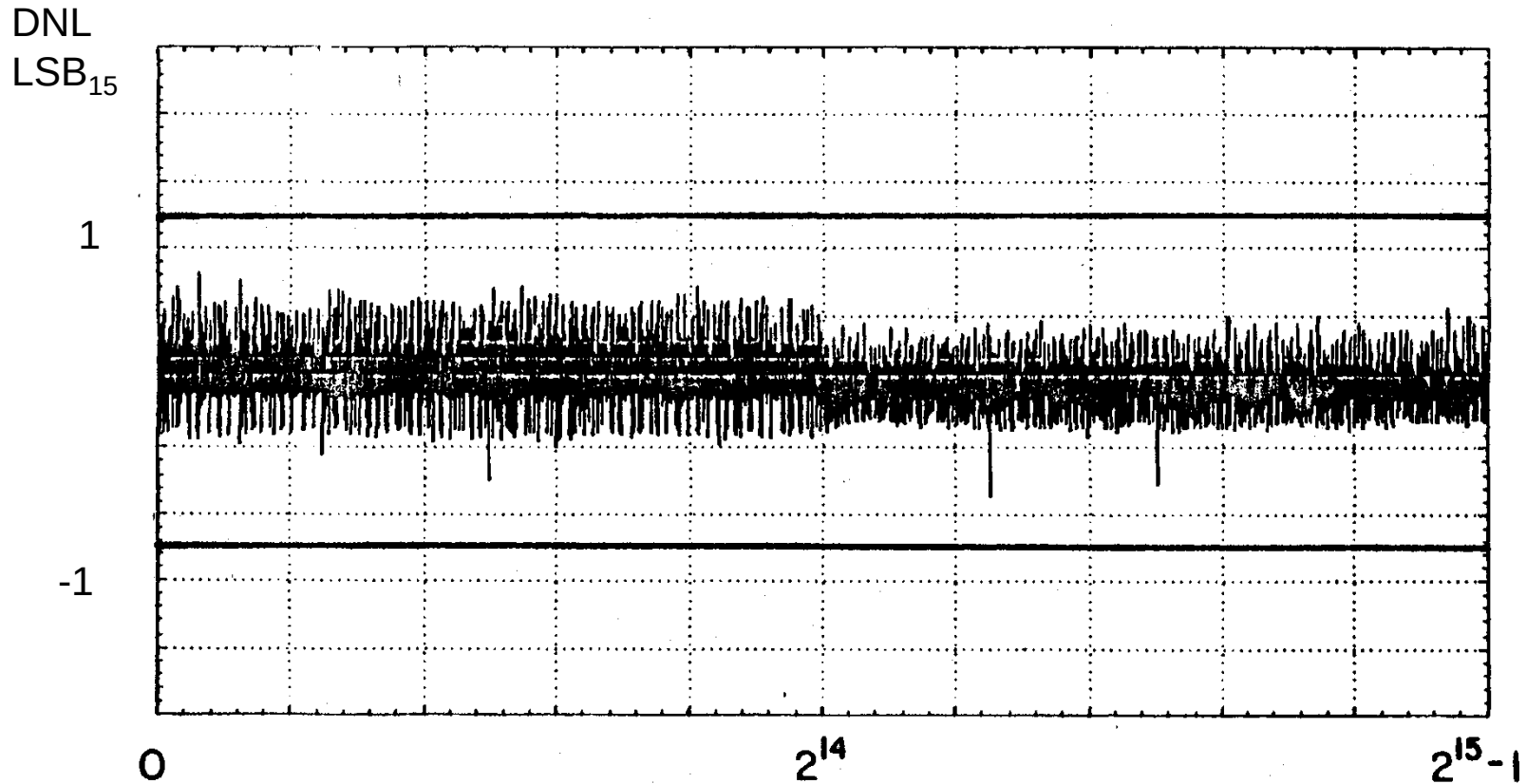
- Open top plate switch, exchange bottom plate voltages
- Use CAL DAC to measure V_{xN}

Initial Capacitor Mismatch



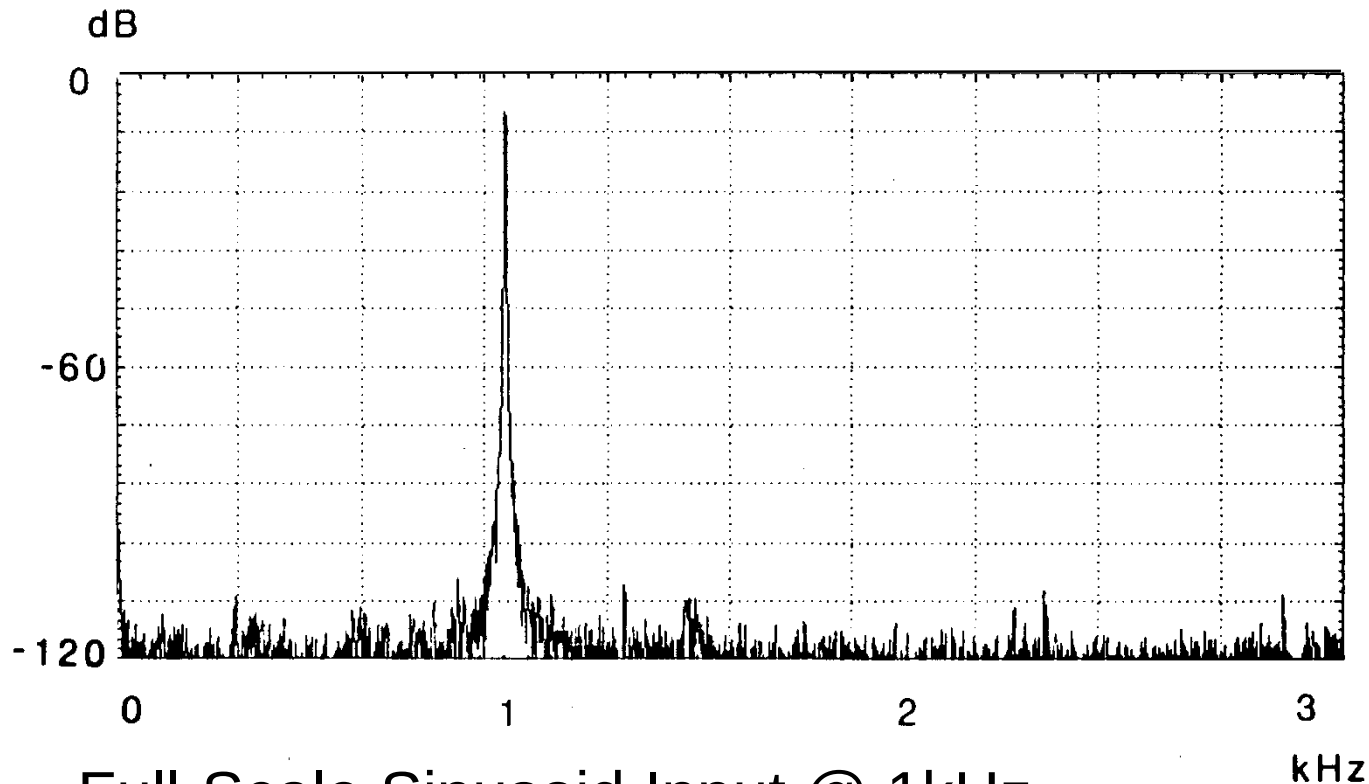
Numbers are LSB_{15}
Corresponds to $\sim 10\text{b}$ initial accuracy

DNL After Calibration



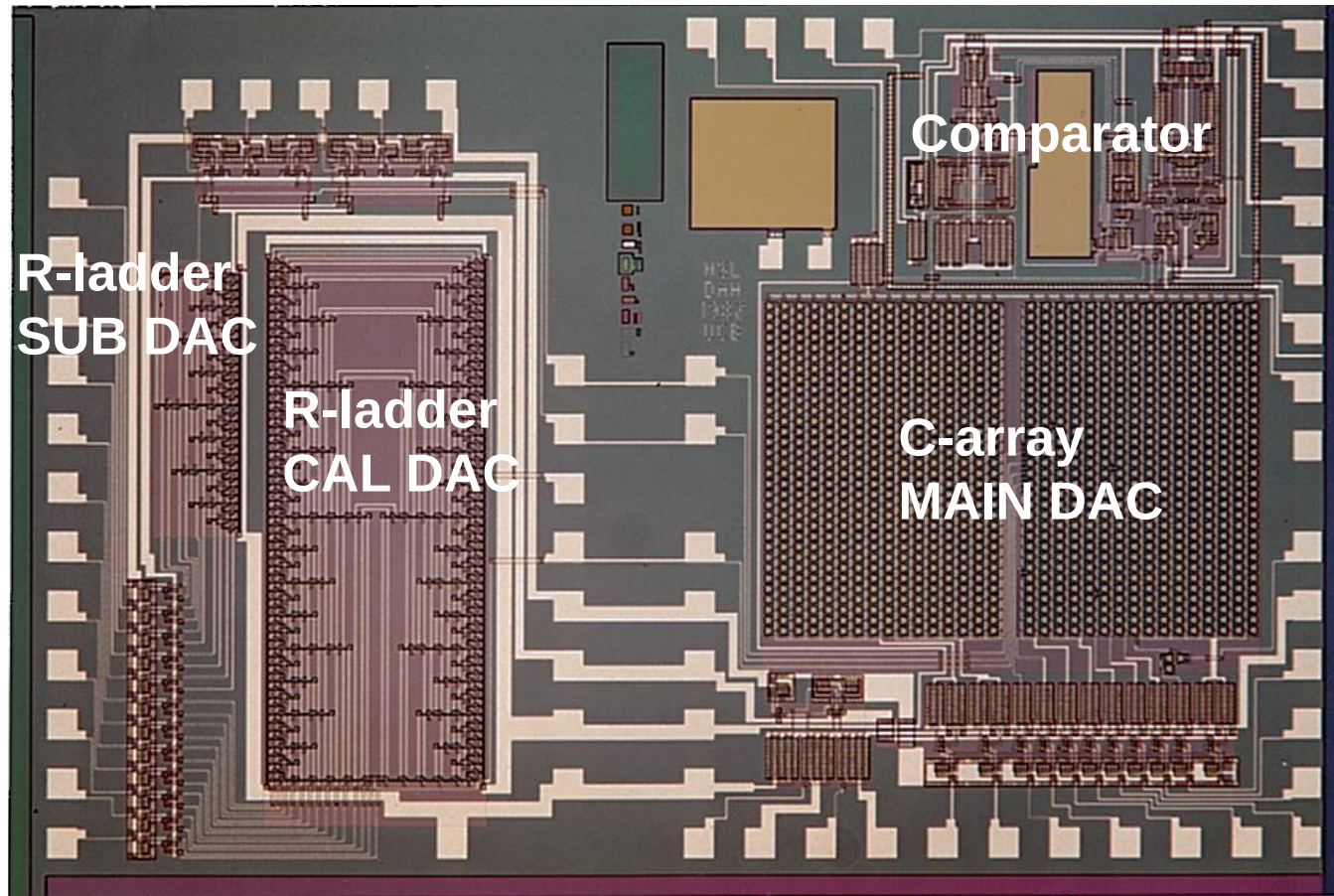
- Max. DNL: $\pm 0.65 \text{ LSB}_{15}$
- Max. INL (Measured directly at major carry points): $\pm 0.8 \text{ LSB}_{15}$

DFT After Calibration



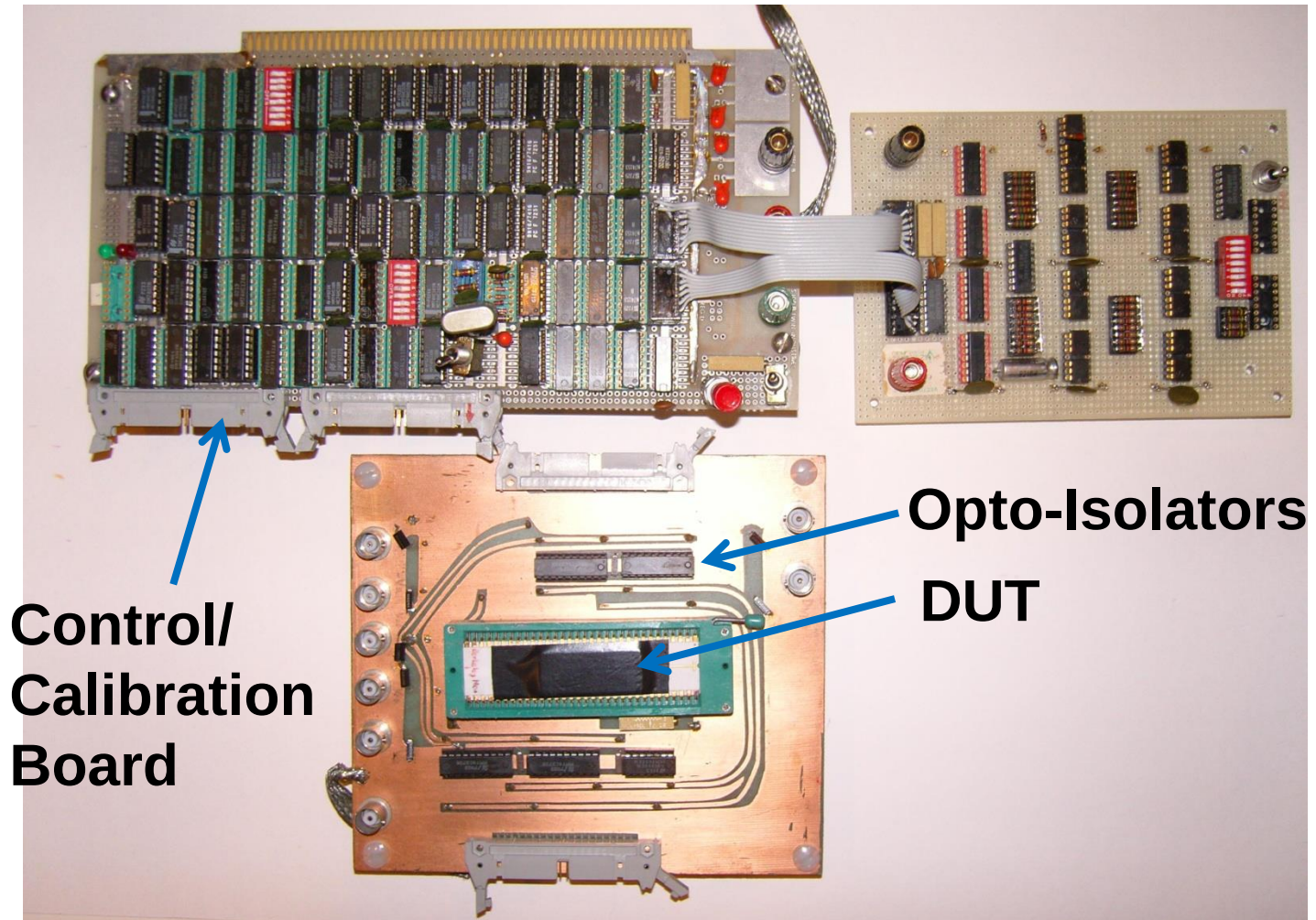
- Full-Scale Sinusoid Input @ 1kHz
- Thermal+1/f Noise Floor = -96dBFS
- Estimated ENOB=14.5

Die Photograph

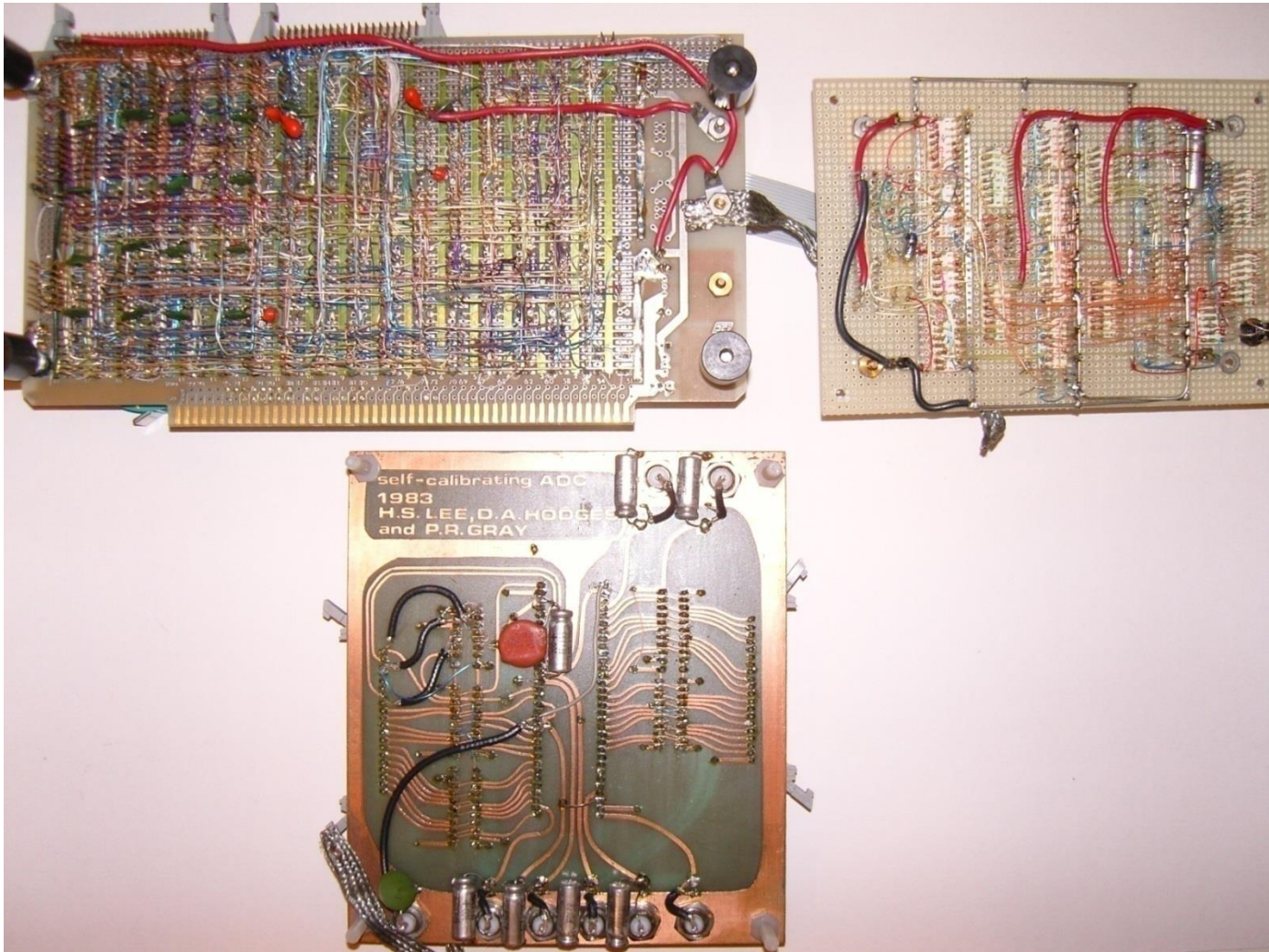


5 μm CMOS, 7.5mm², FOM=35pJ/step

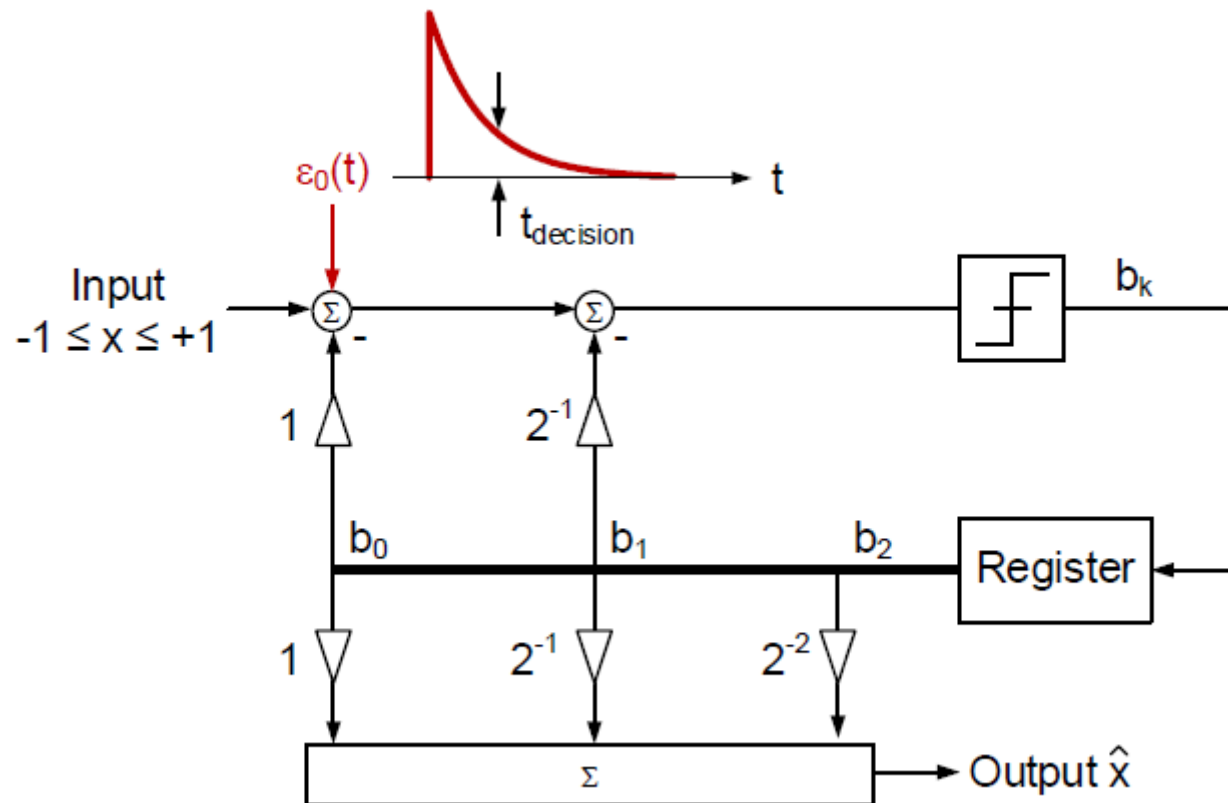
Test Setup (1)



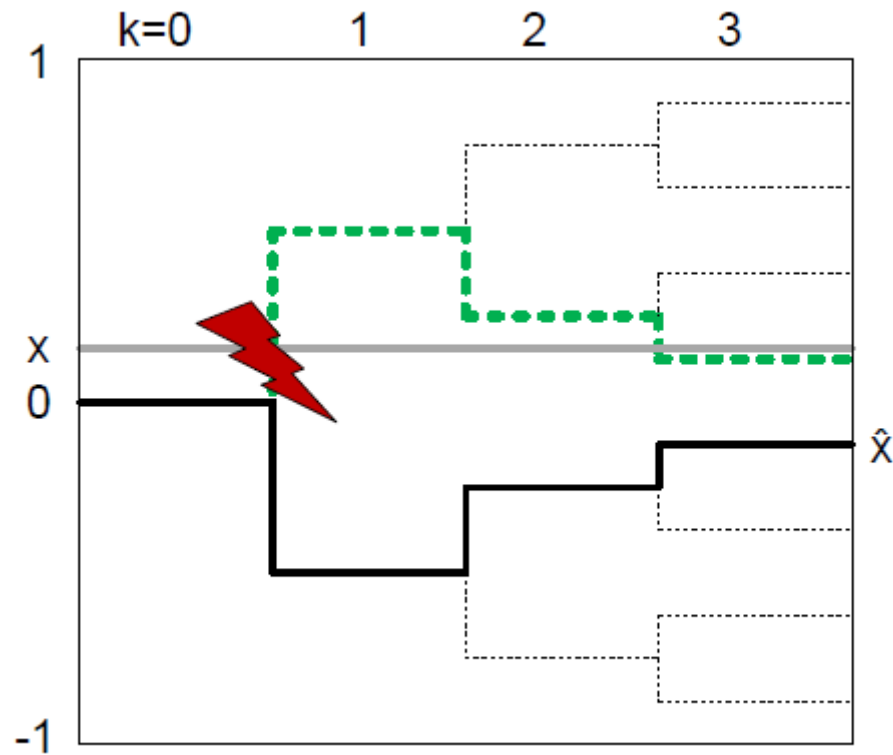
Test Setup (2)



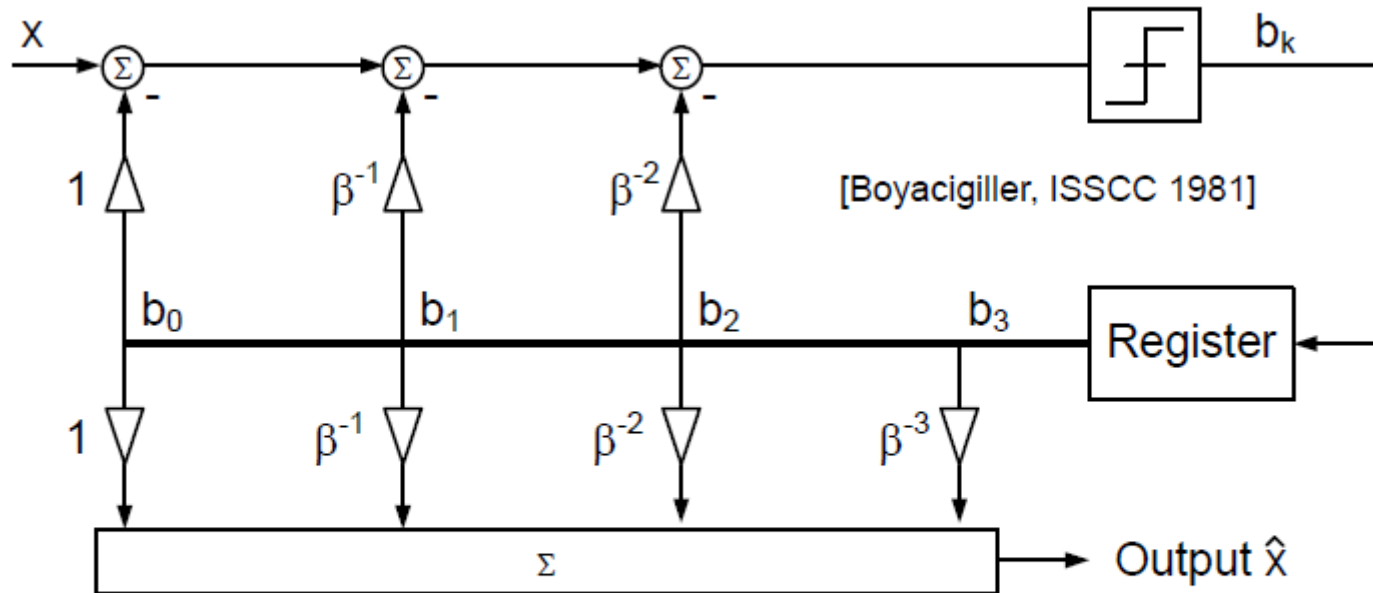
DAC Settling Errors



Error in the MSB Decision

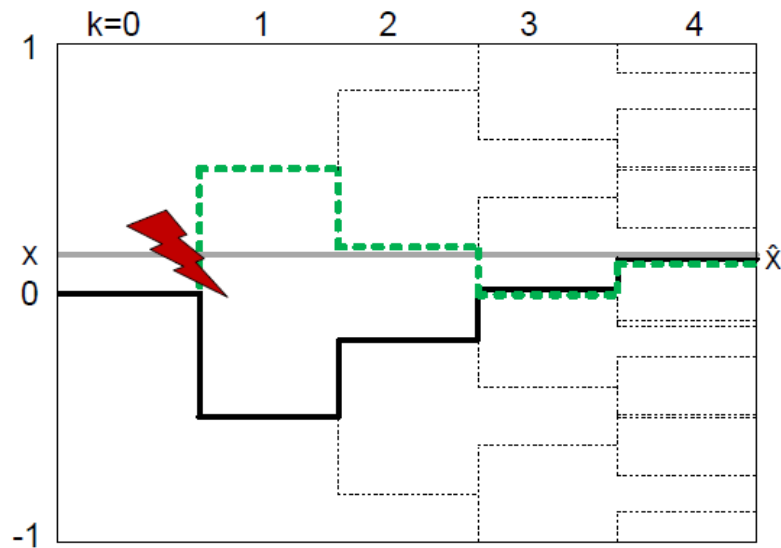


With redundancy (radix < 2)



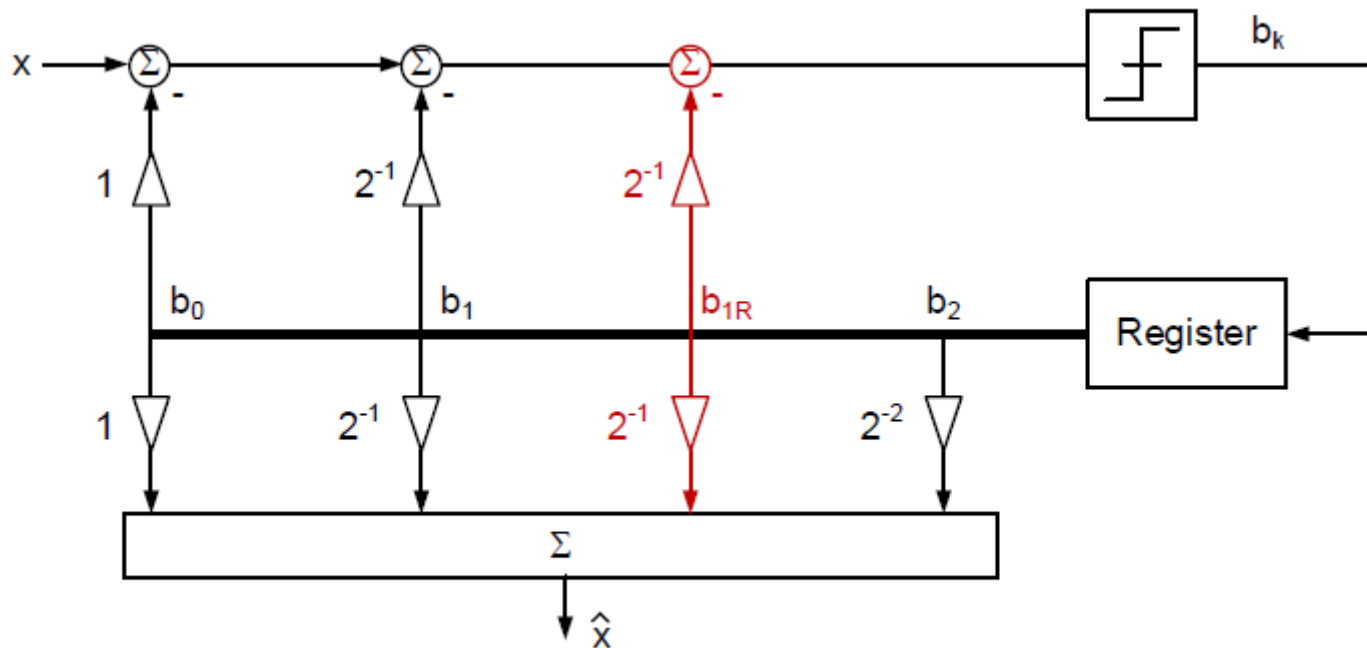
$$\beta < 2 \quad \sum_{i=1}^{B-1} \beta^{-i} > 1 \quad (\text{Key property})$$

Error in the MSB Decision ($\beta=2^{0.7}$)



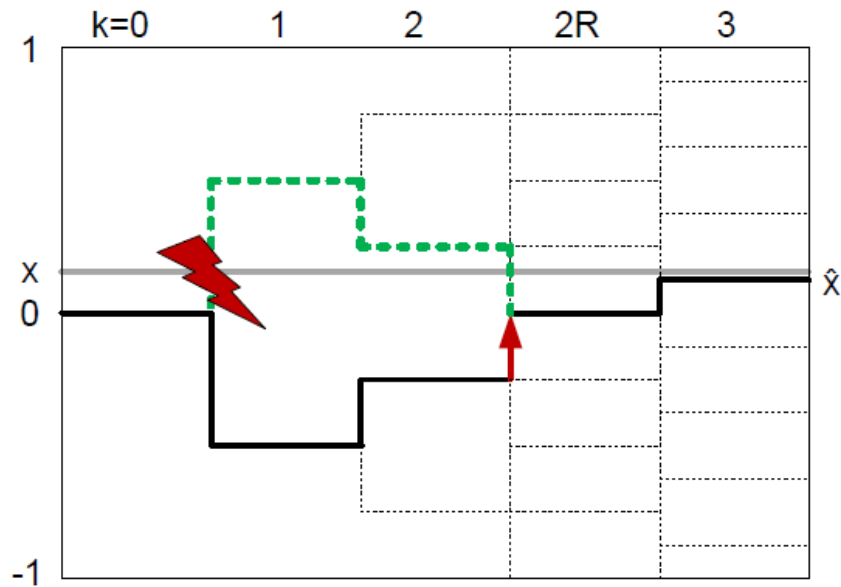
- Green line:
 - 1,0,0,1
 - $D_{\text{out}} = \beta^{-1} + \beta^{-4}$
- Black line:
 - 0,1,1,1
 - $D_{\text{out}} = \beta^{-2} + \beta^{-3} + \beta^{-4}$

SAR ADC with Redundant Step



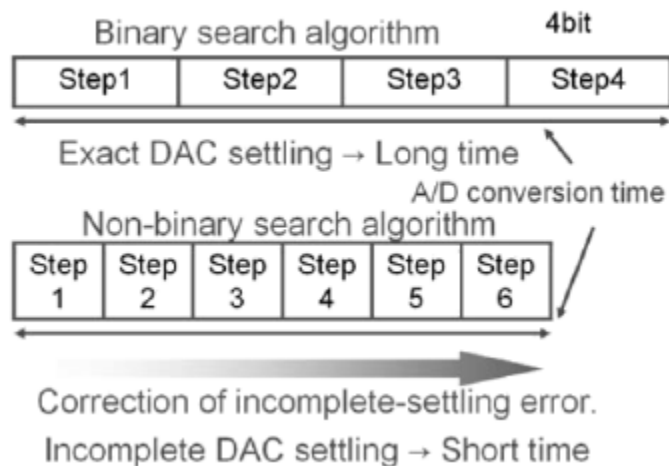
[C.-C. Liu, ISSCC 2010]

Error in the First Decision



- Green line:
 - 1,0,0,1
 - $D_{\text{out}} = 2^{-1} + 2^{-3}$
- Black line:
 - 0,1,1,1
 - $D_{\text{out}} = 2^{-2} + 2^{-2} + 2^{-3}$

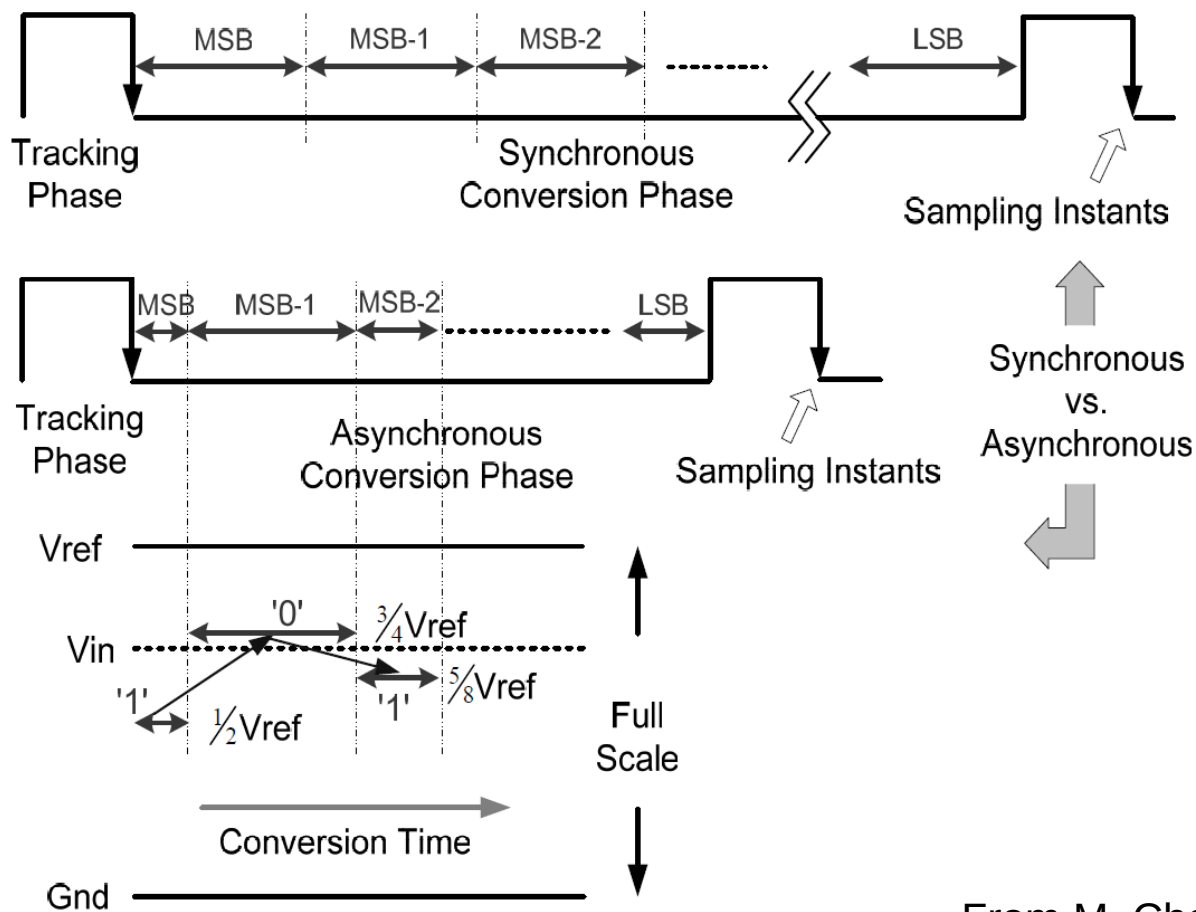
Potential Speed Benefit



Algorithm	Binary	Prior	Generalized
Time slot for each step	6.3τ	2.2τ	1.9τ
Number of steps	10	12	12
Total conversion time	56.7τ	24.2τ	20.9τ

T. Ogawa et al., IEICE Trans. on Fundamentals of Electronics, Communications and Computer Sciences, Feb. 2010.

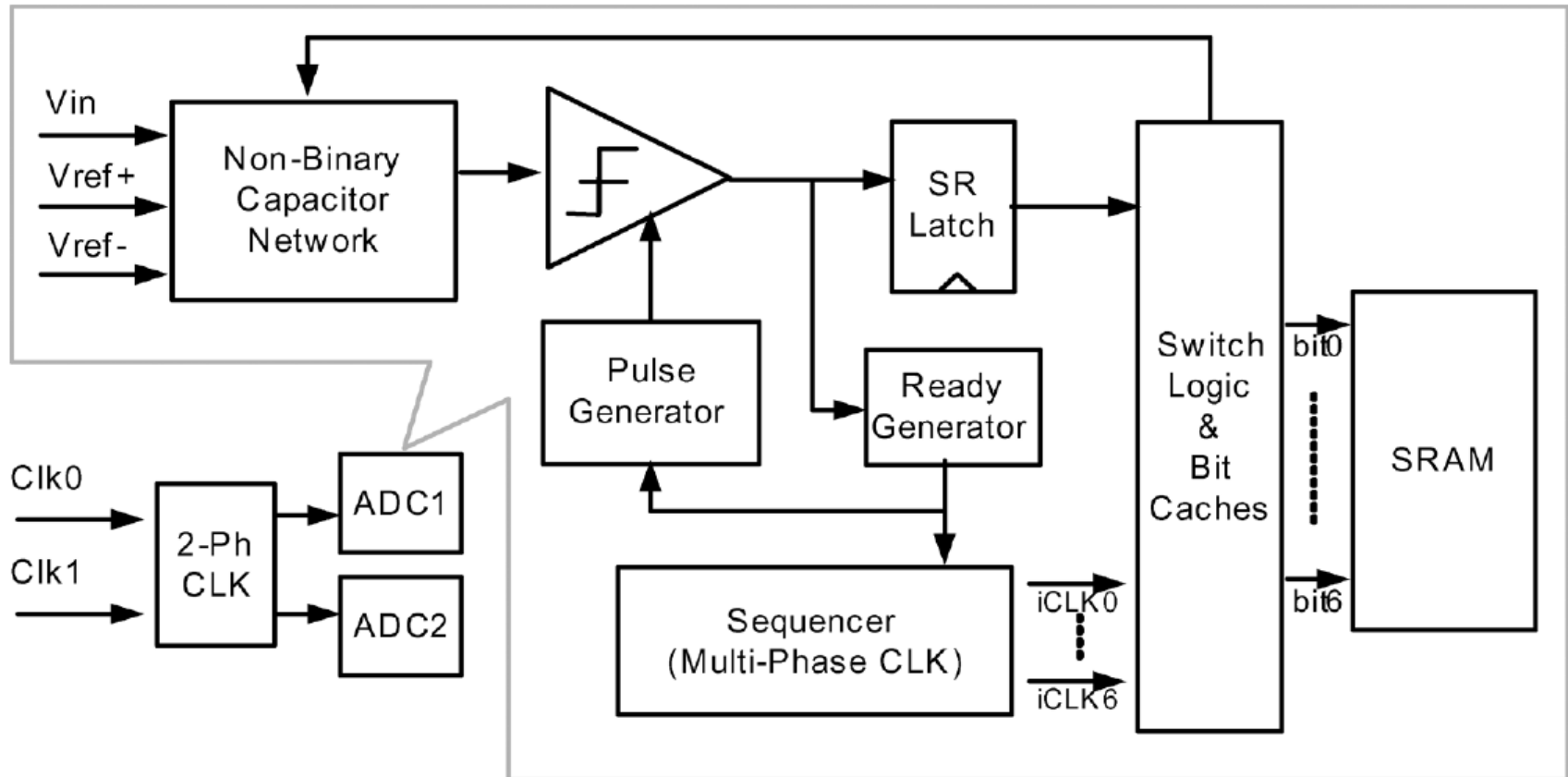
Asynchronous SAR



From M. Chen, ISSCC, 2006

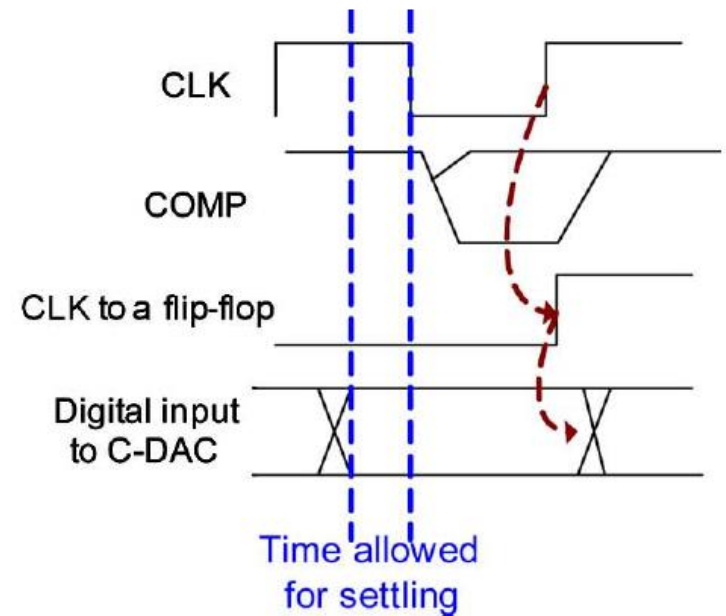
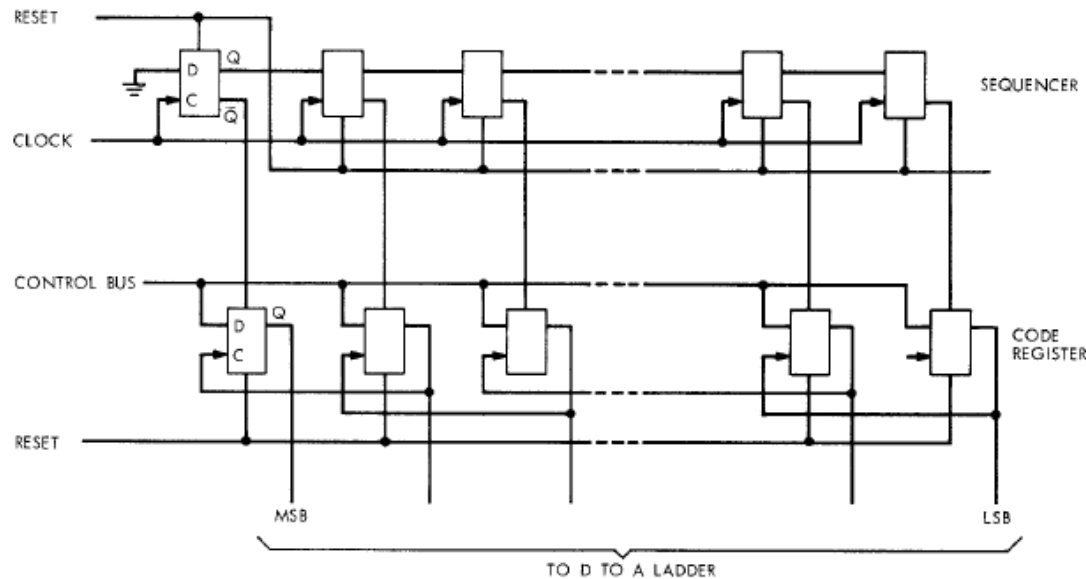
Architecture

From M. Chen, ISSCC, 2006

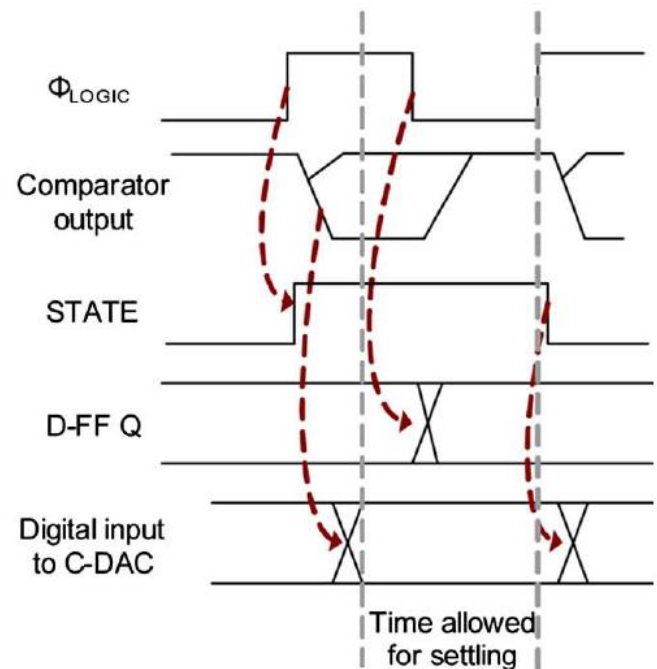
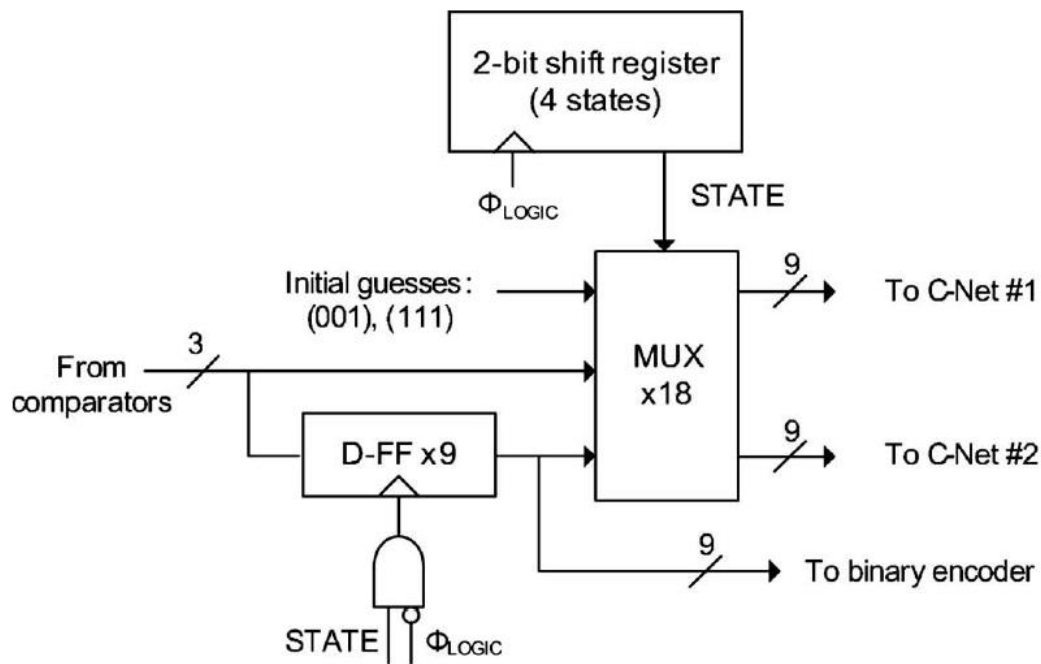


A NAND gate is sufficient as a ready generator

Typical SAR Logic and Timing Diagram

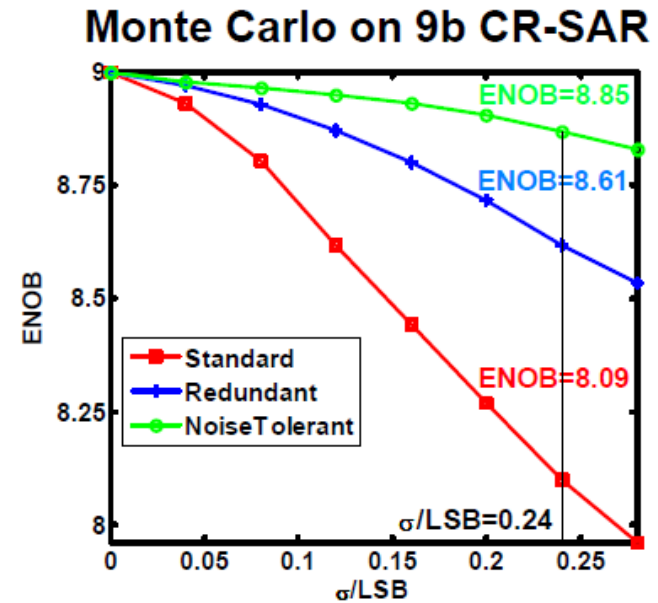
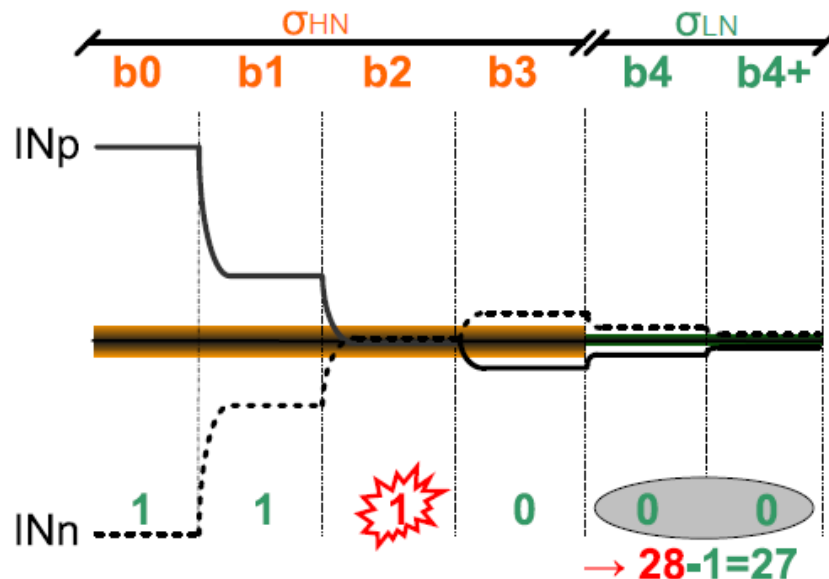


Fast SAR logic



Comparator noise reduction (1)

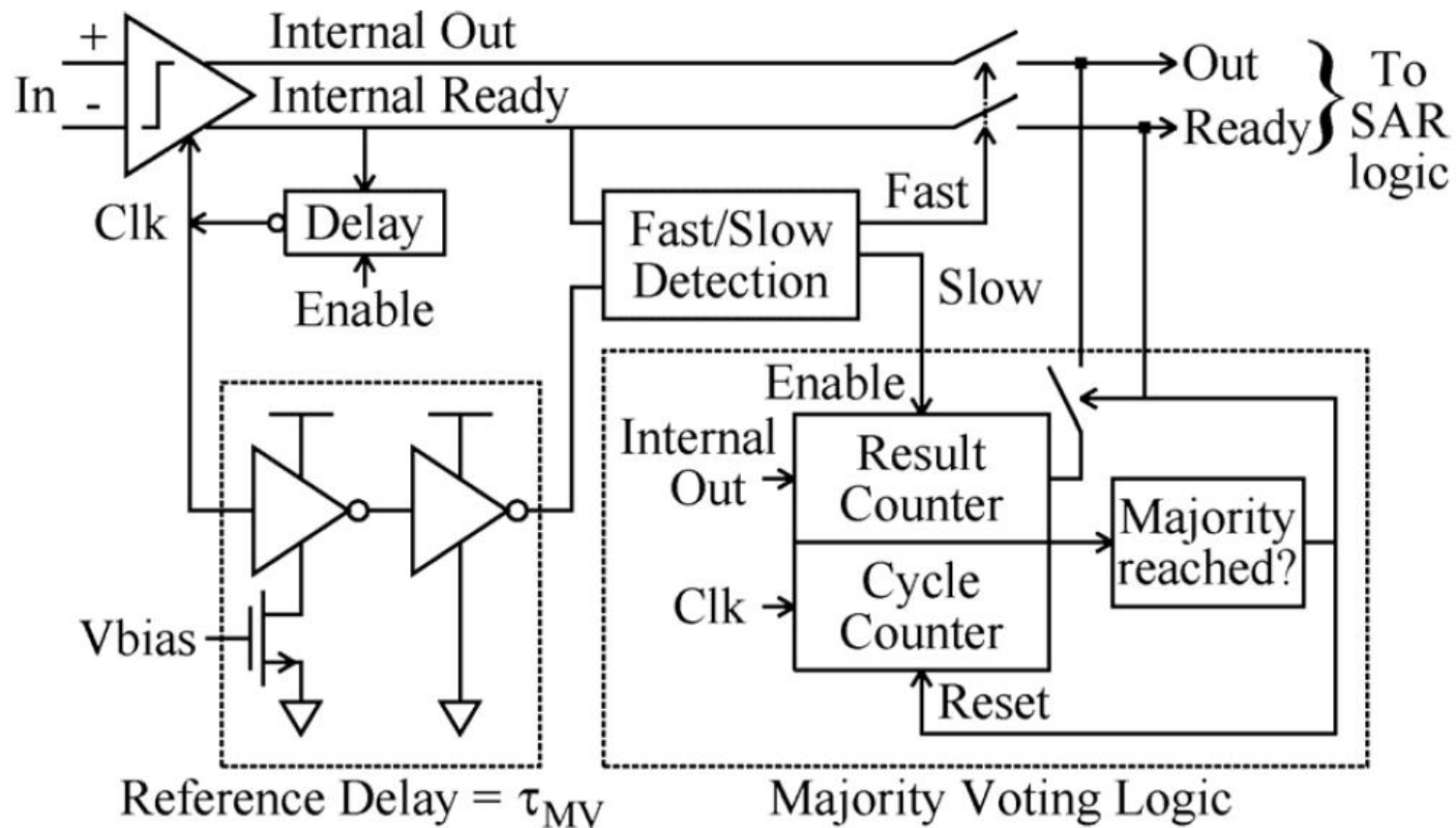
1b redundancy + noise reconfigurable comparator



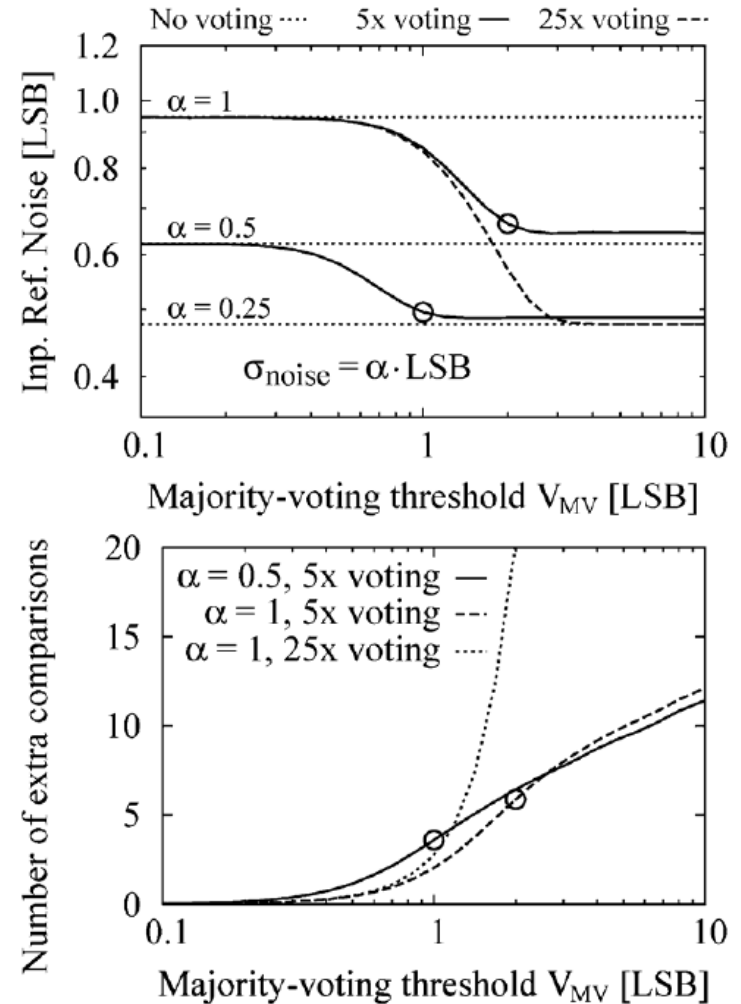
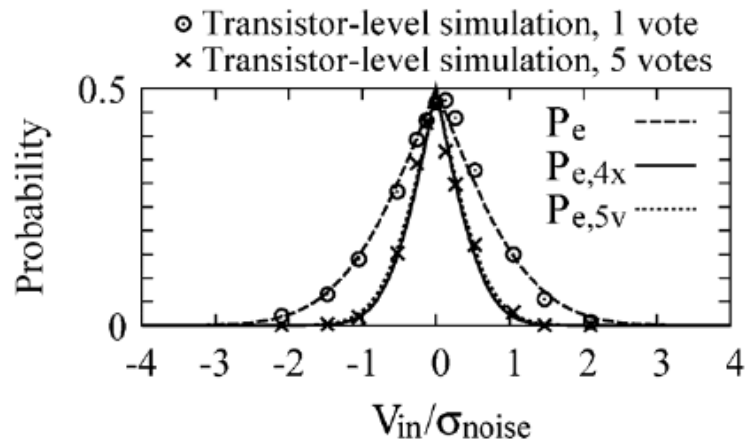
1. (N-1) low power comparisons @ σ_{HN}
2. N and N+1 high power comparisons @ σ_{LN}

Comparator noise reduction (2)

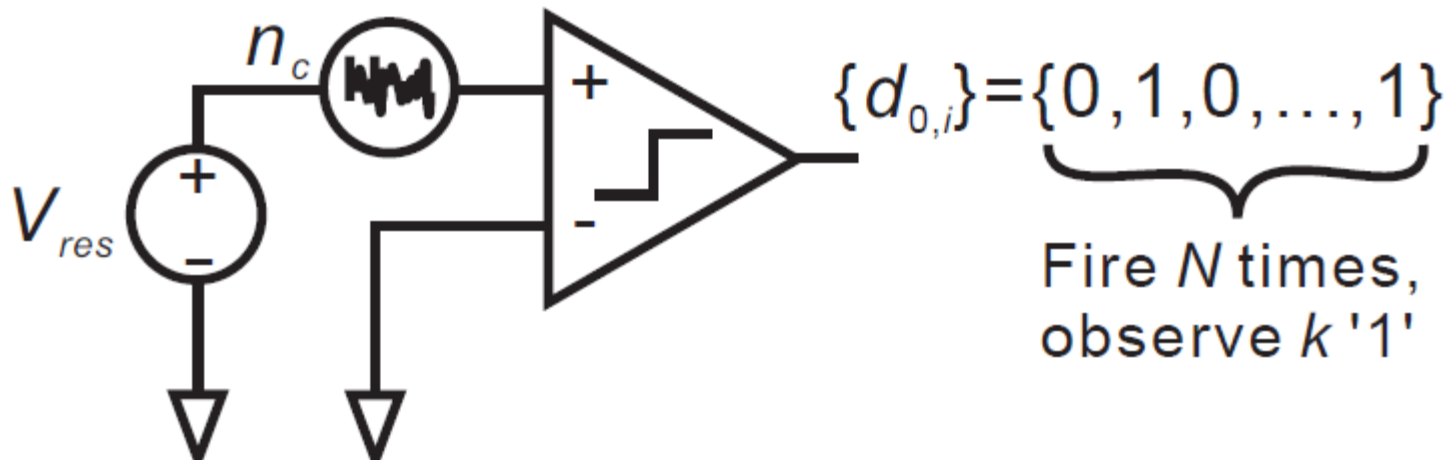
Majority voting



Comparator noise reduction (3)

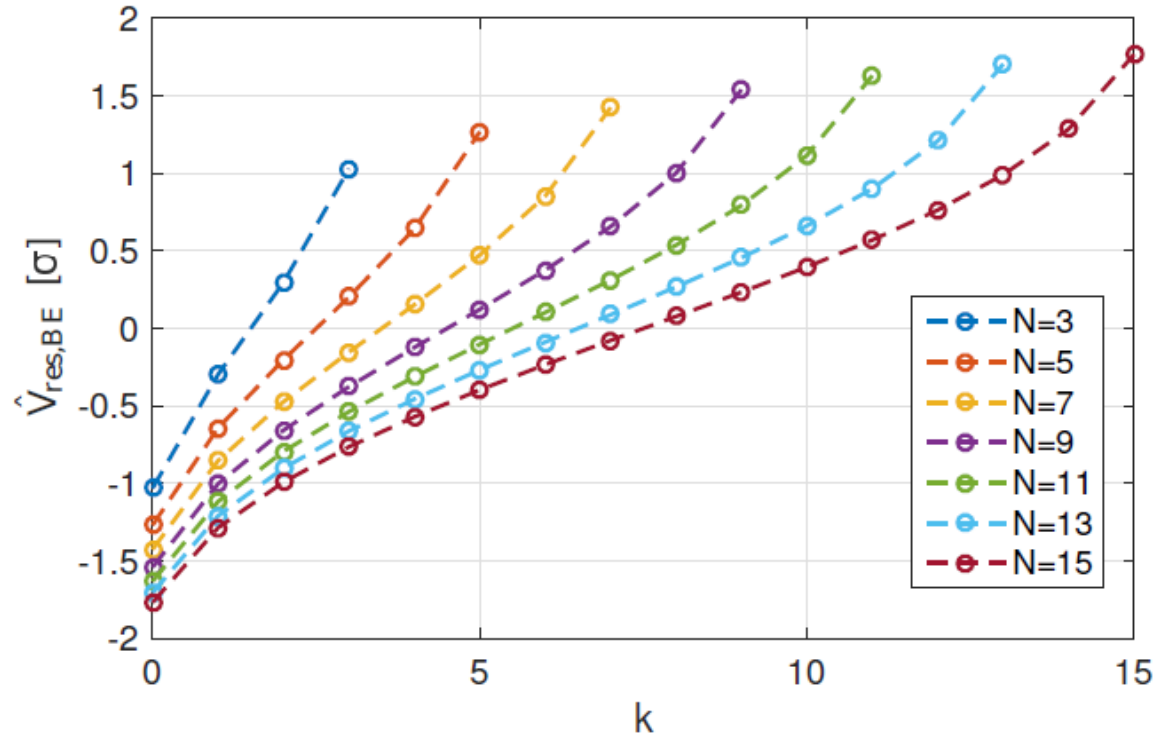


Proposed statistical estimation



We can estimate V_{res} from k .
This is enabled by the noise!

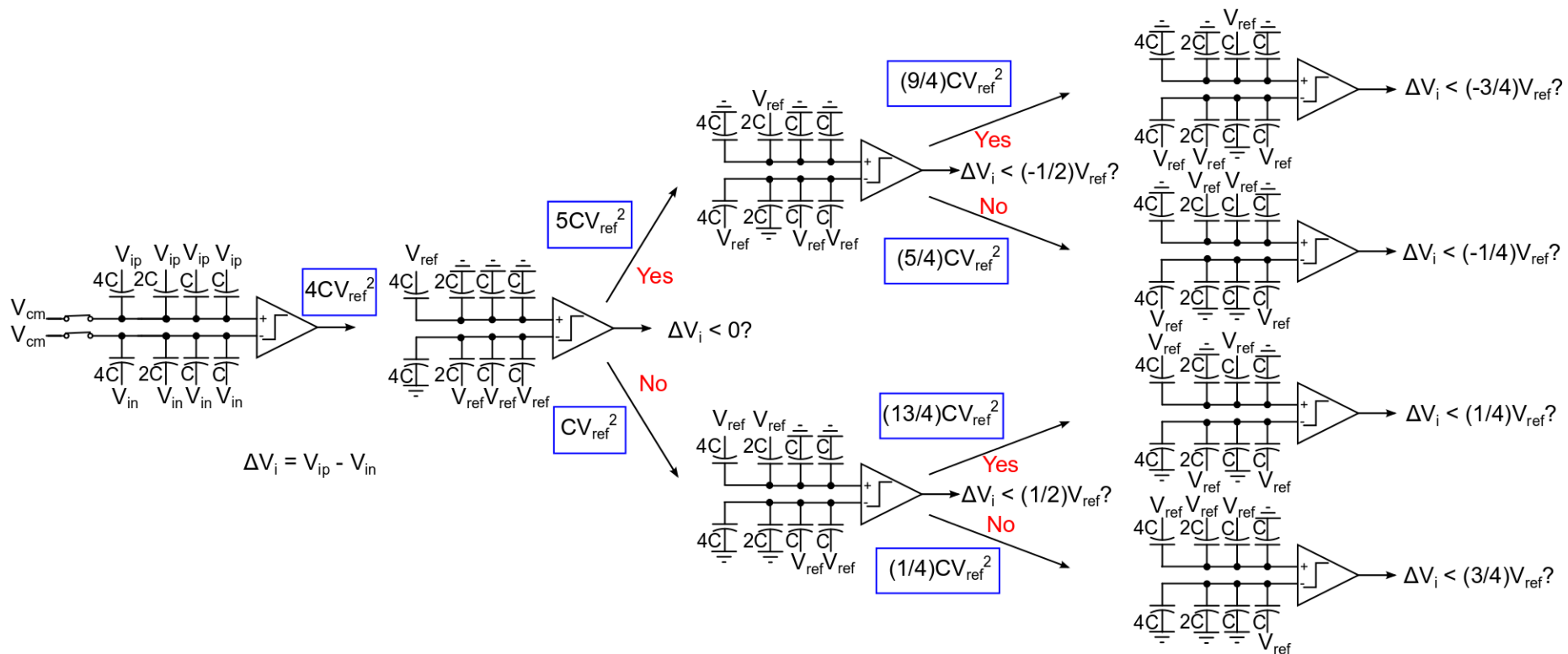
Bayes estimator



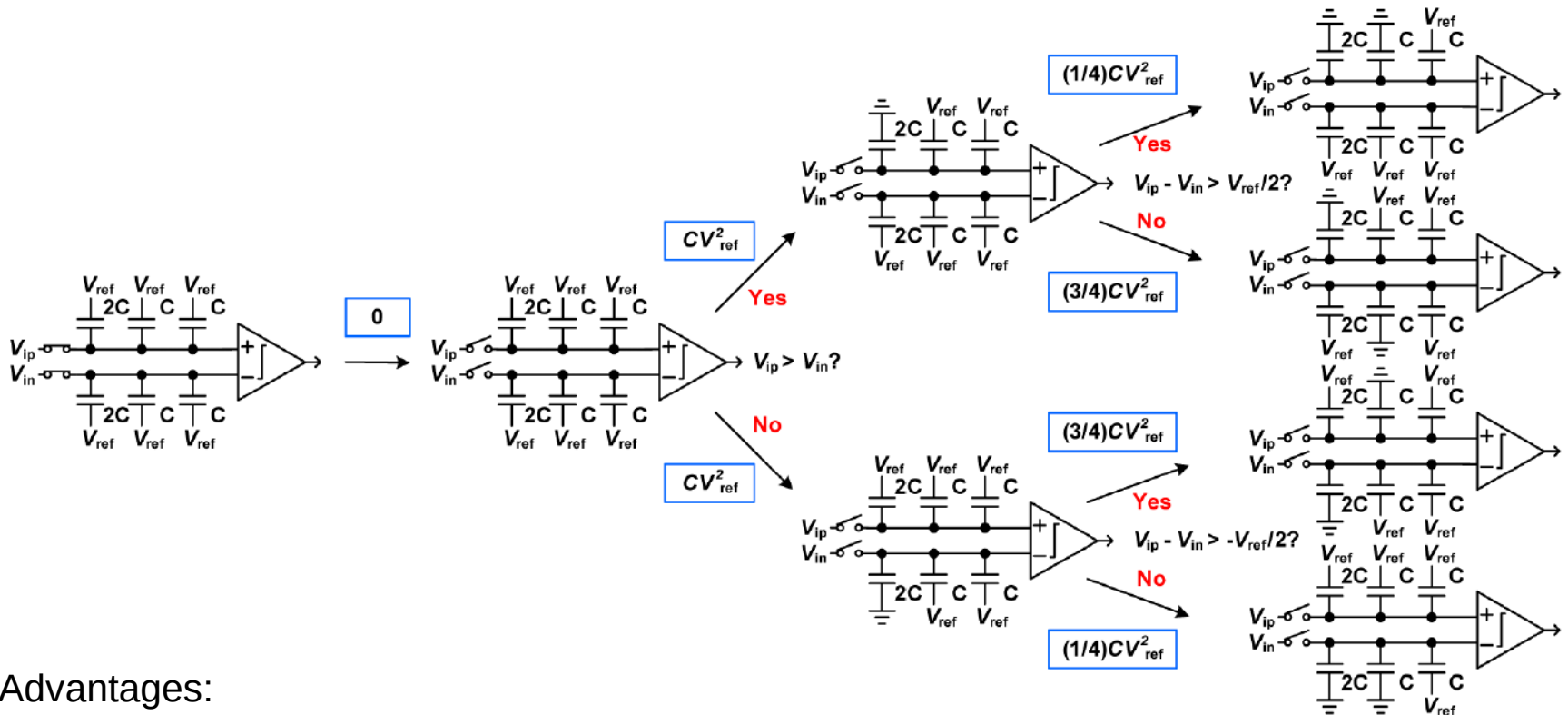
It achieves the lowest estimation error

$$MSE(k) = E \left[(V_{res} - \hat{V}_{res})^2 | k \right] = \left(\hat{V}_{res} - E(V_{res} | k) \right)^2 + Var(V_{res}^2 | k)$$

Conventional switching technique



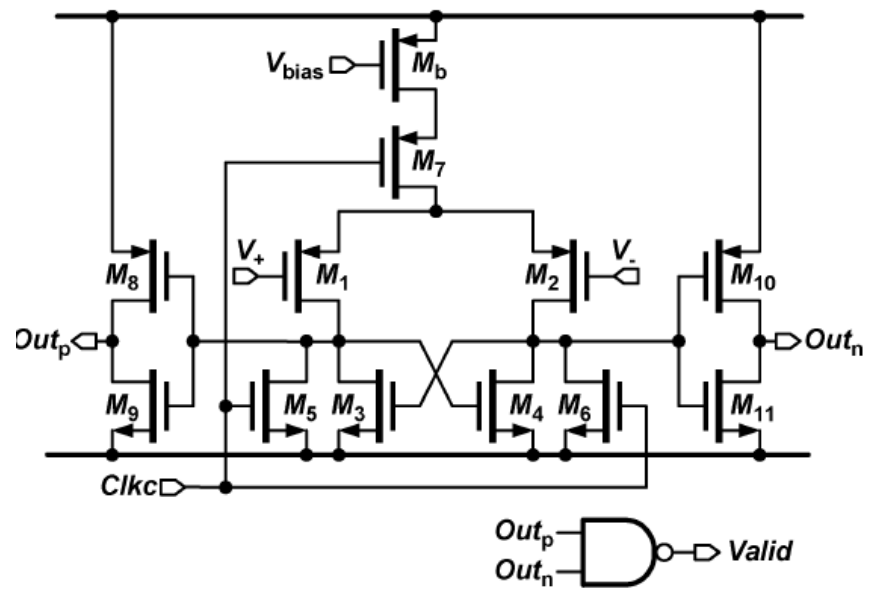
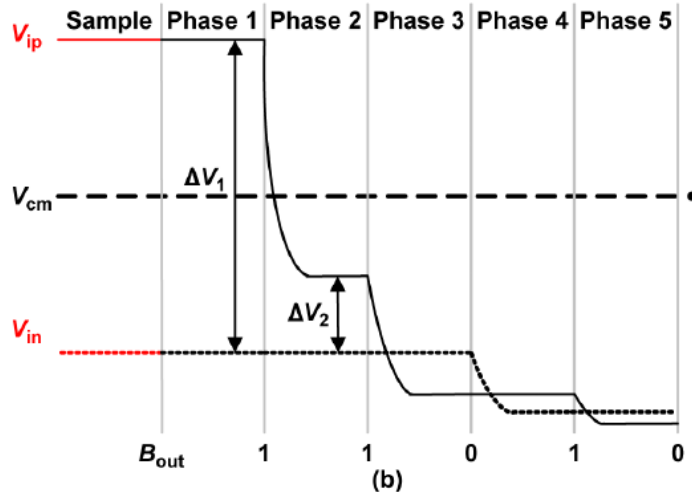
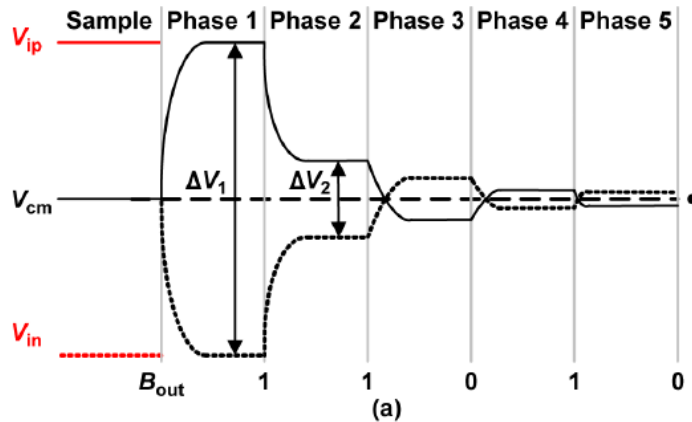
- Need $16C$ for 3-b
- Maximum 4 switching events per bit
- Large DAC reference power



Advantages:

- 1) 2x reduction in # unit capacitors
- 2) 81% reduction in DAC reference power
- 3) Only 1 switching every cycle, low switching power

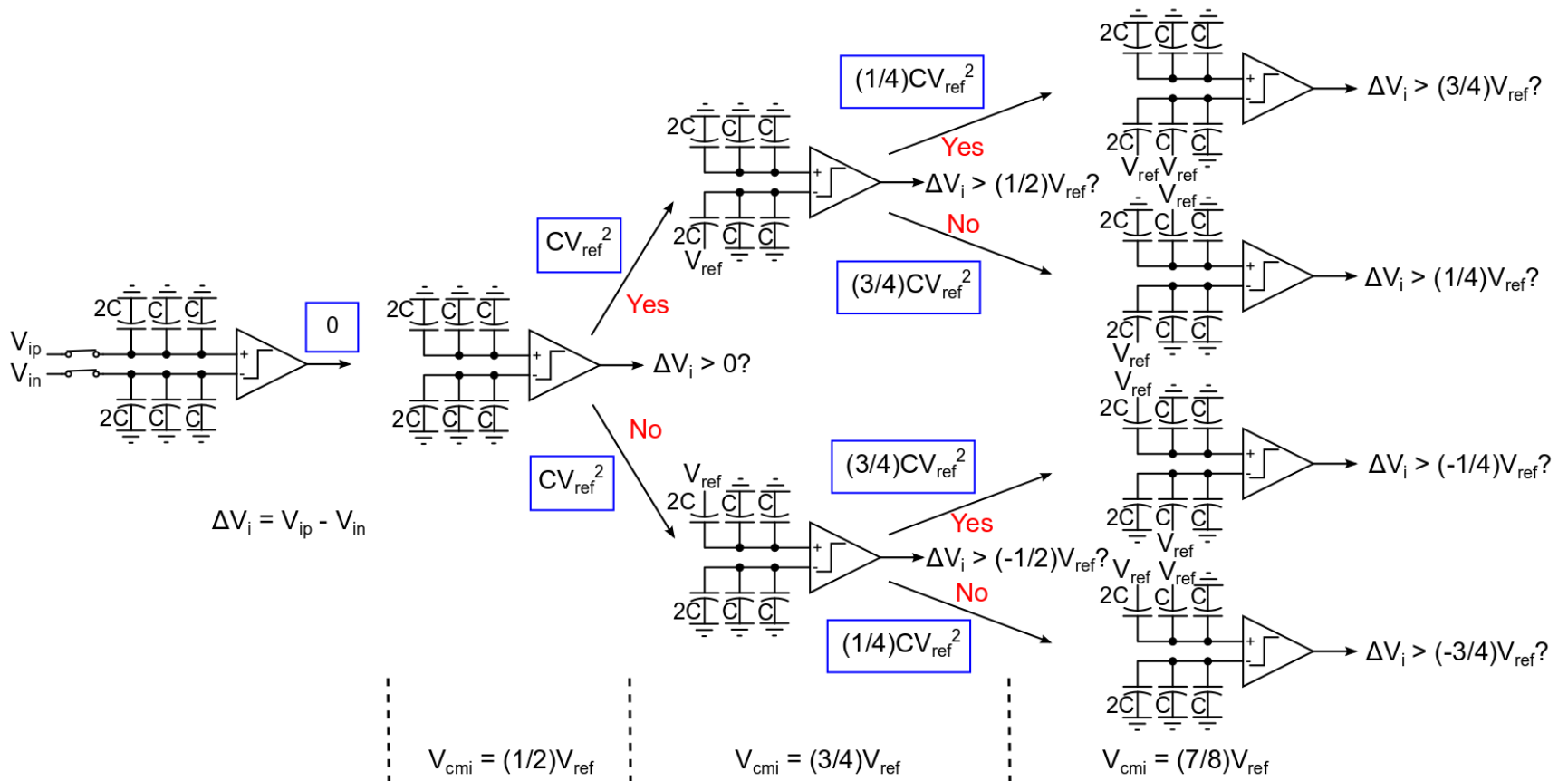
Common-mode variation



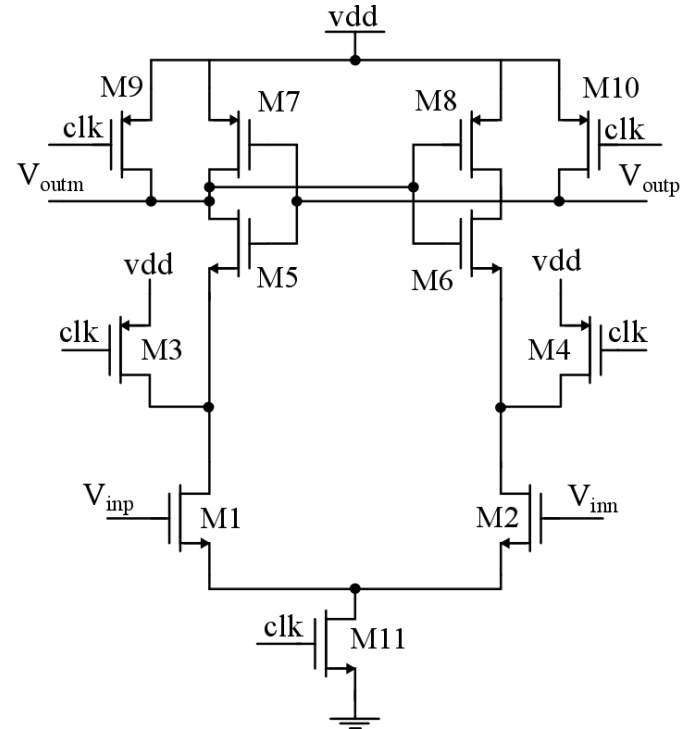
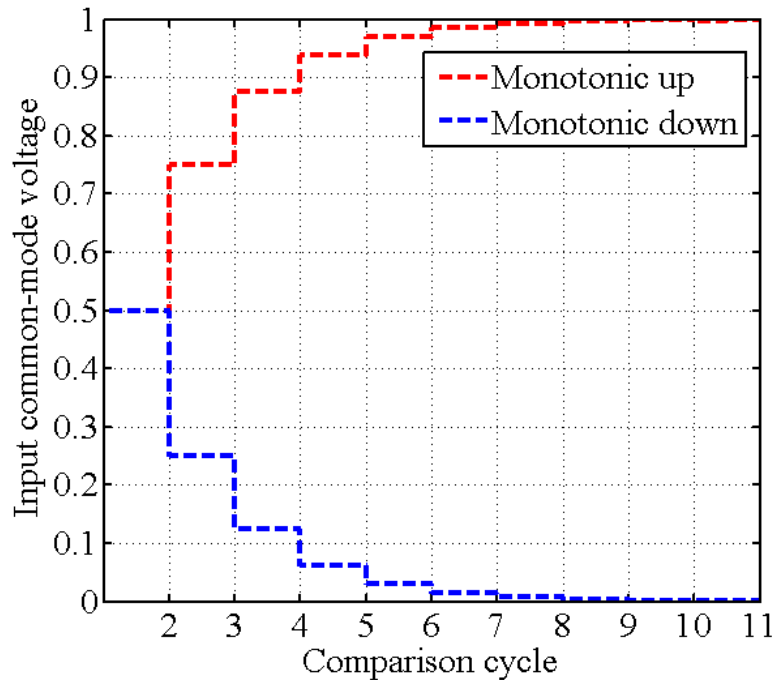
Disadvantages:

- 1) V_{cm} decreases to 0
- 2) PMOS input pair (smaller f_T)
- 3) Dissipate static power

Monotonic switching up (1)



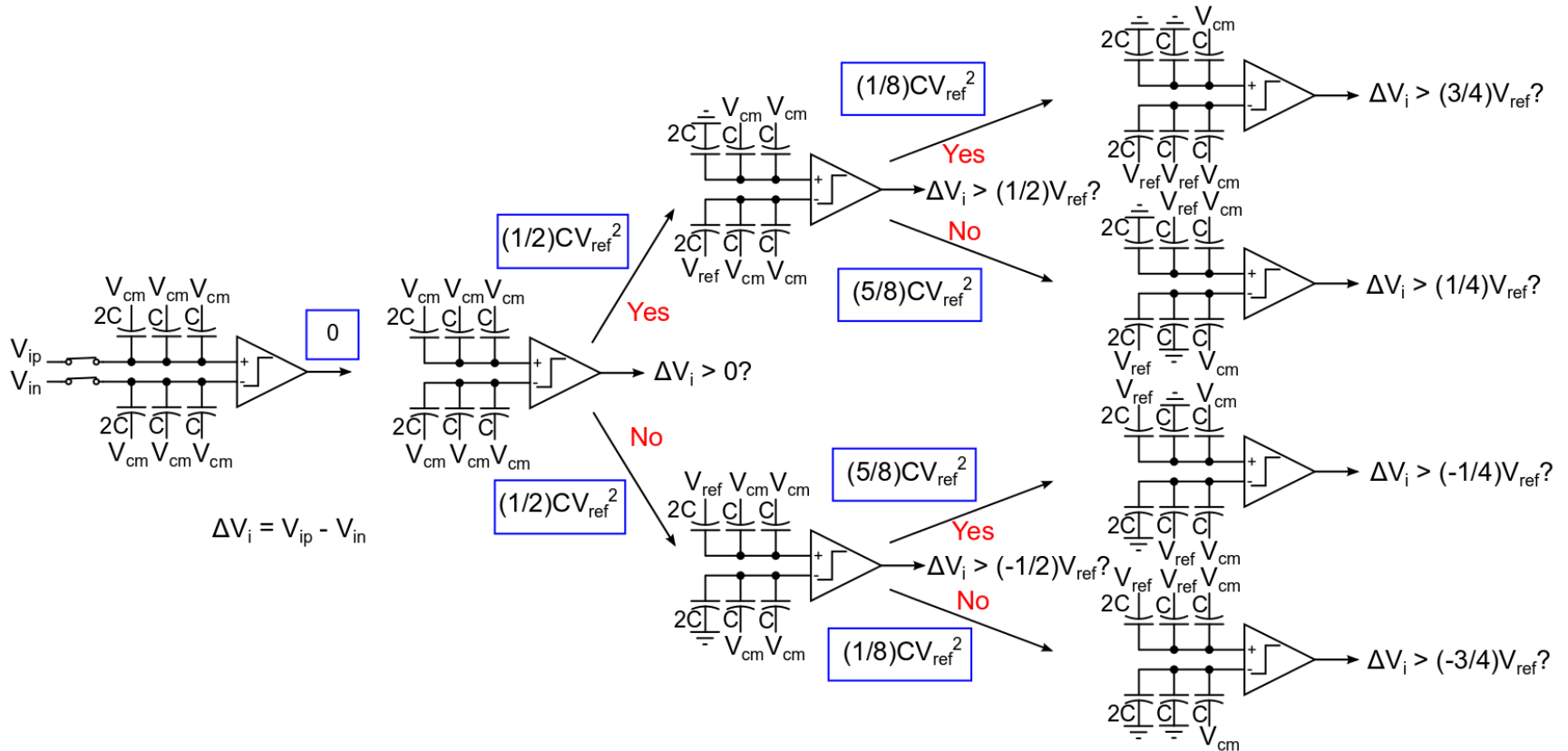
Monotonic switching up (2)



Disadvantages:

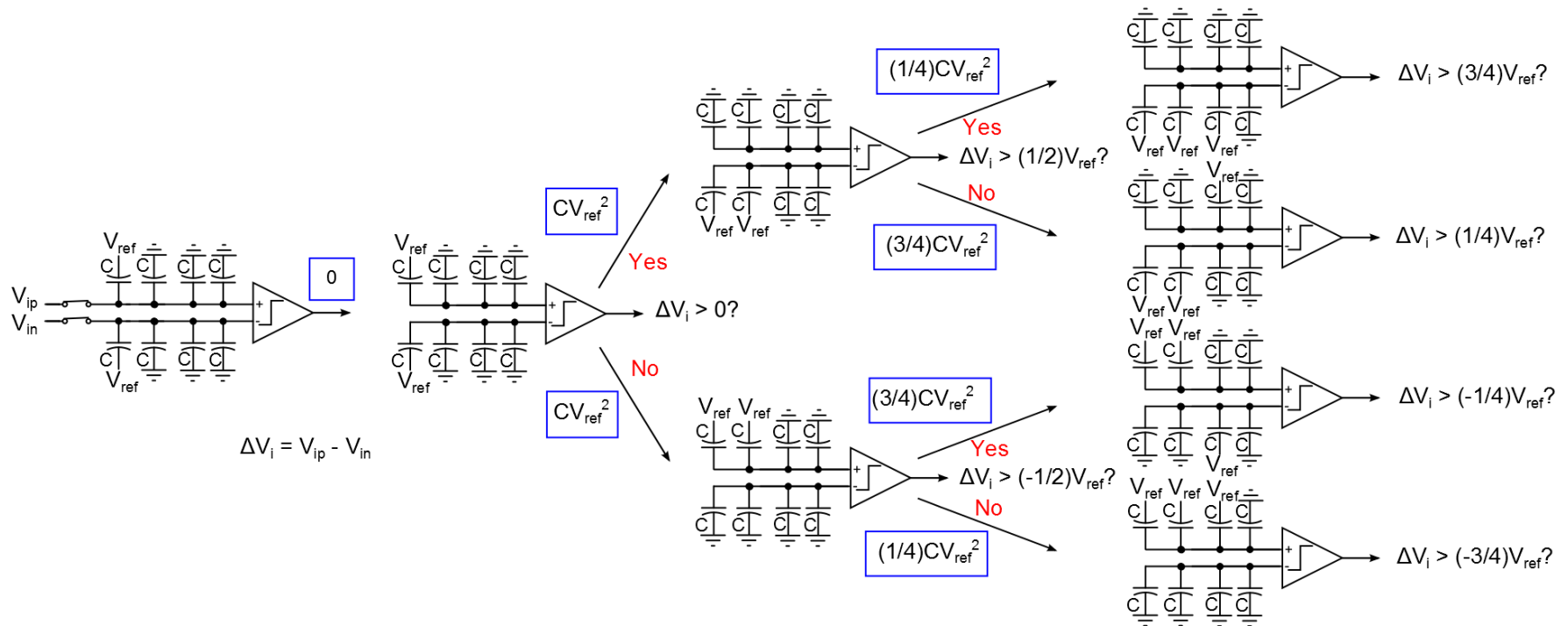
- 1) V_{cm} goes to V_{dd}
- 2) Large comparator offset
- 3) Large comparator noise

V_{cm} -based switching technique



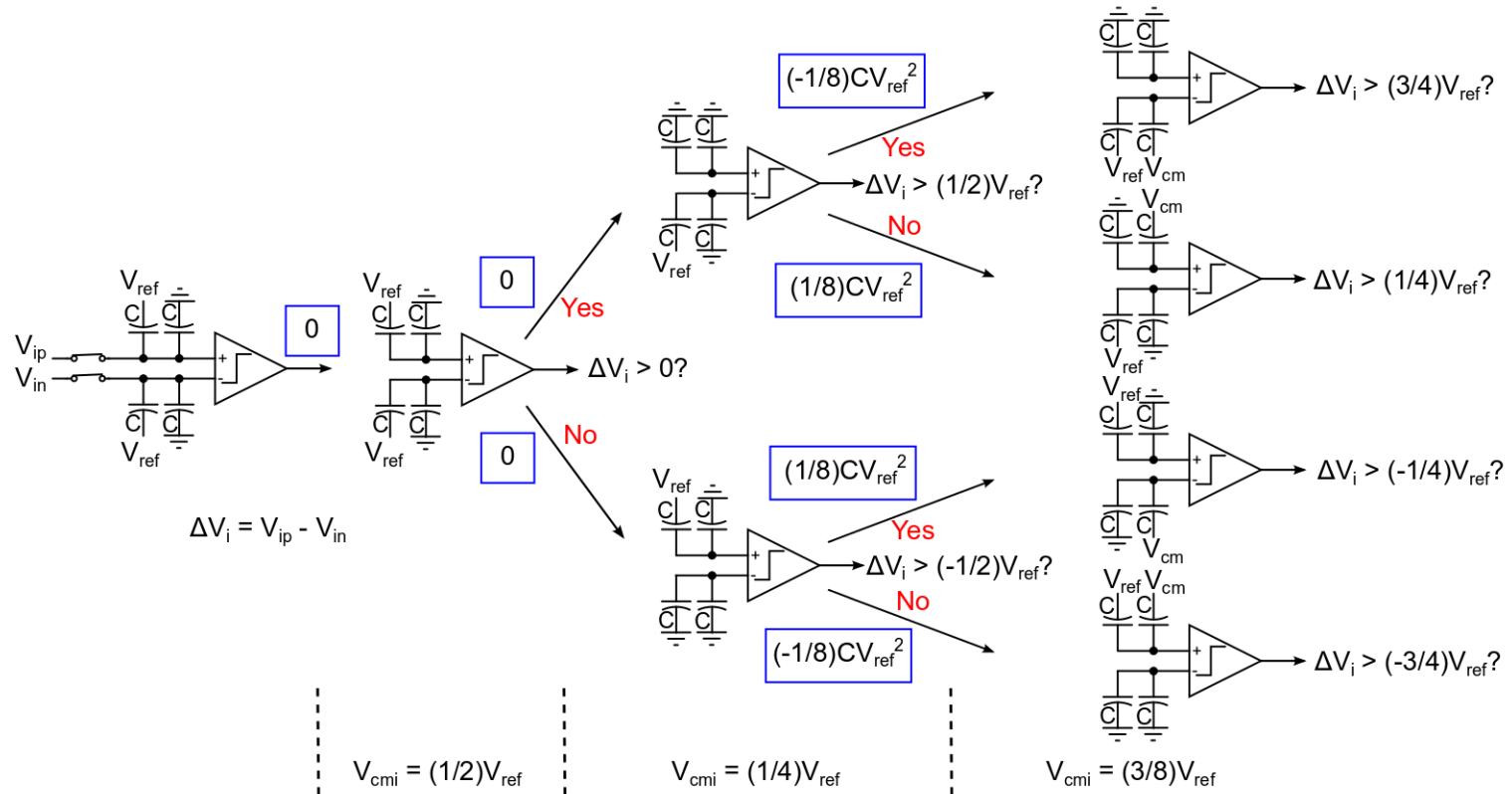
- Need low-impedance V_{cm} reference
- Increased switch power

Split MSB capacitor



- Increased hardware complexity
- Additional capacitor mismatch

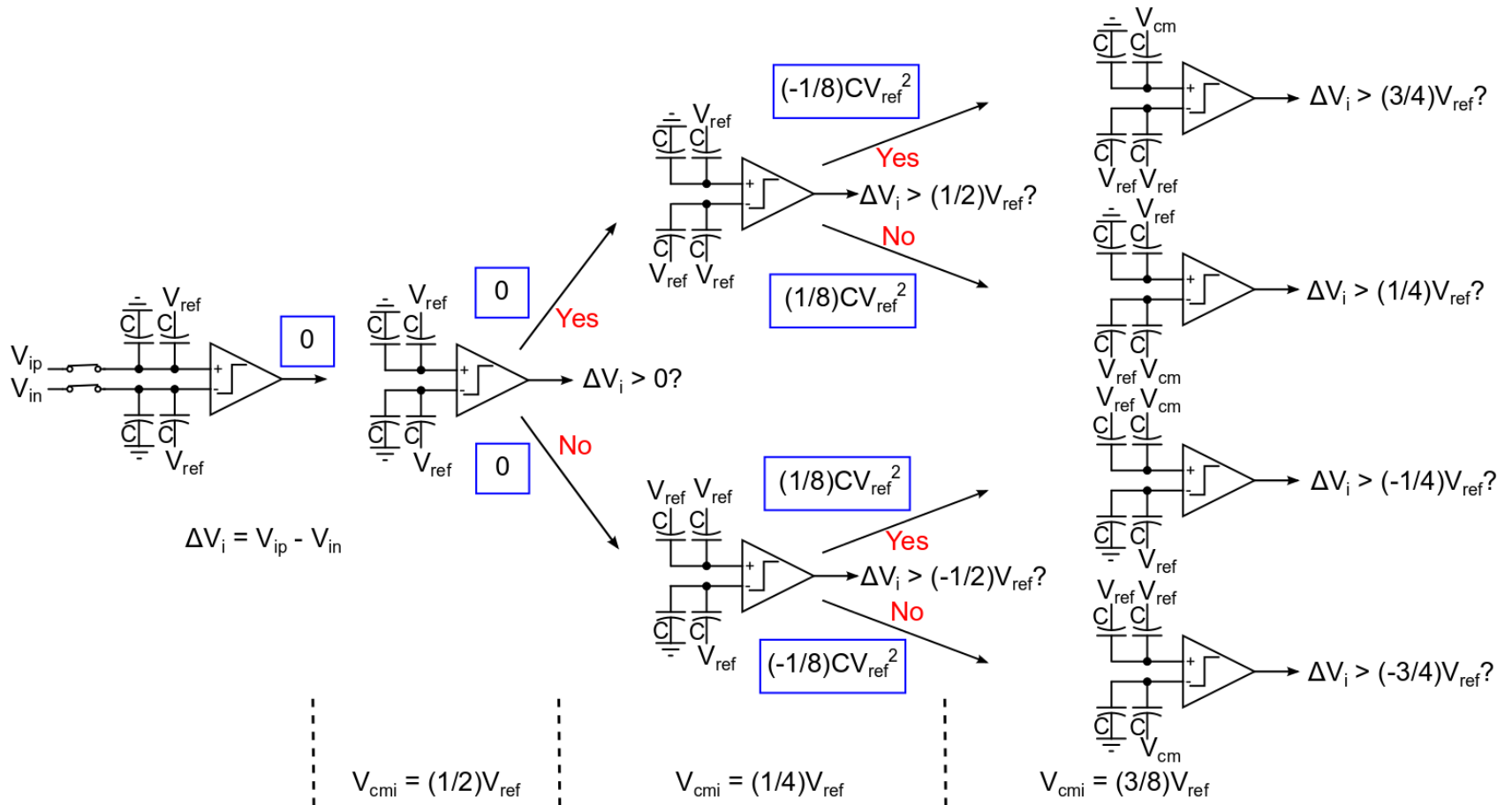
Bidirectional single-side switching



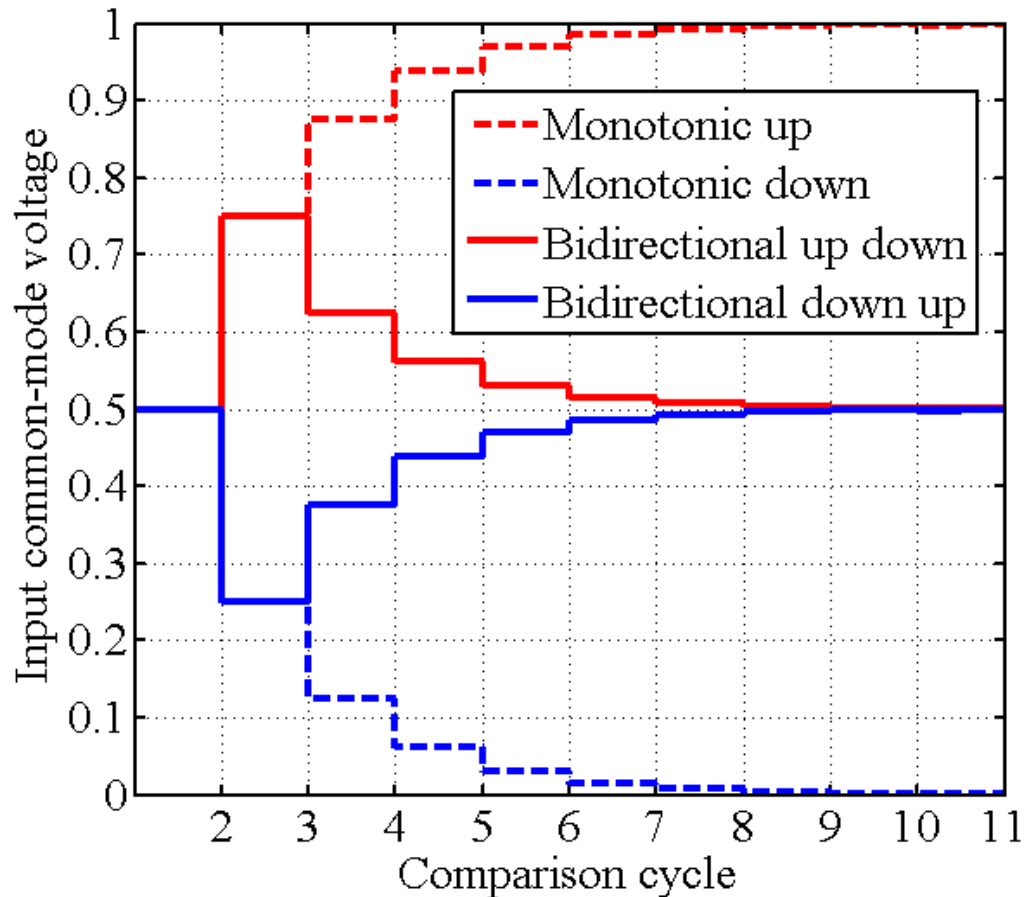
Advantages:

- 1) 95% reduction in DAC reference power
- 2) 4X reduction in # unit capacitor

Bidirectional up-down

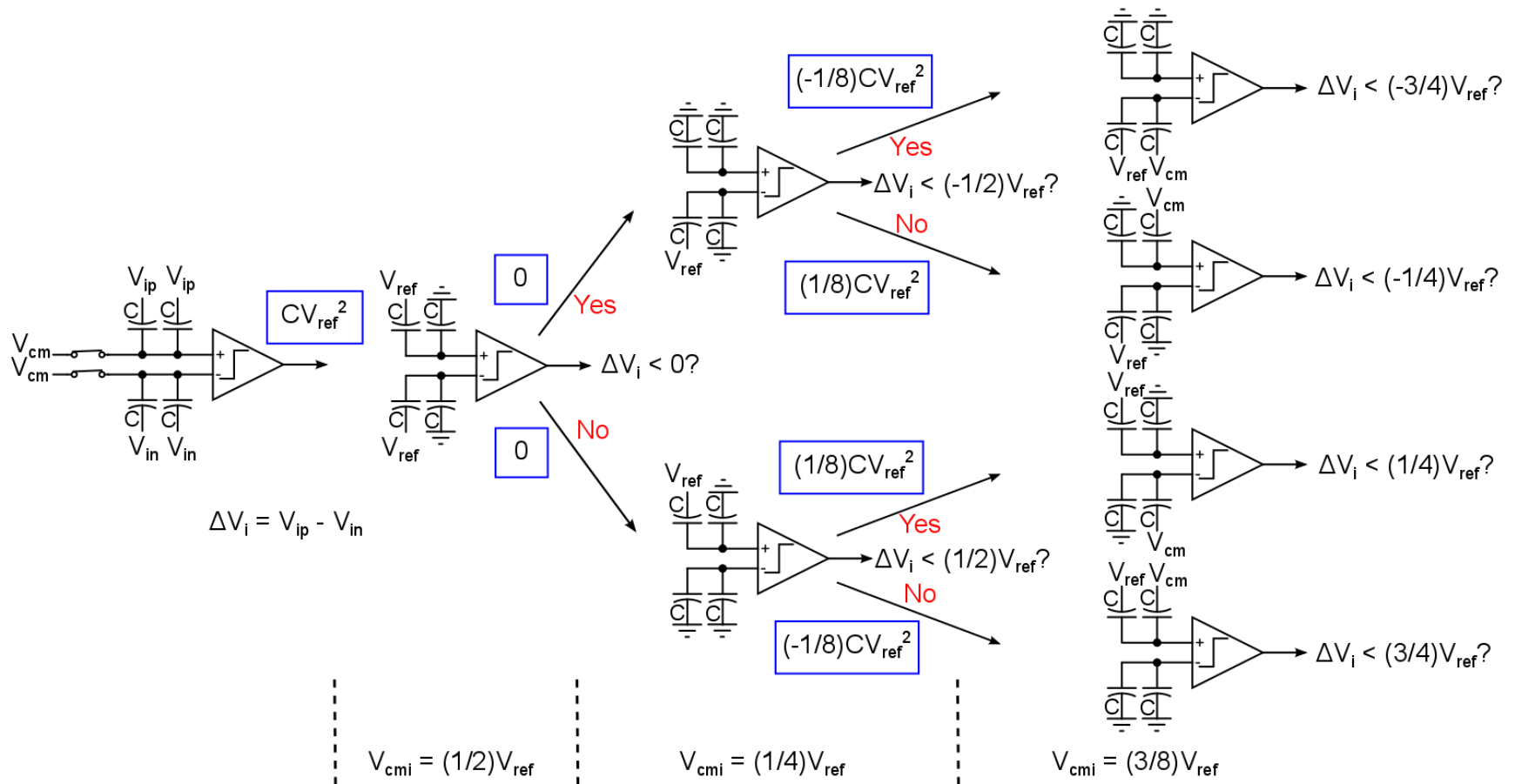


Common-mode variation

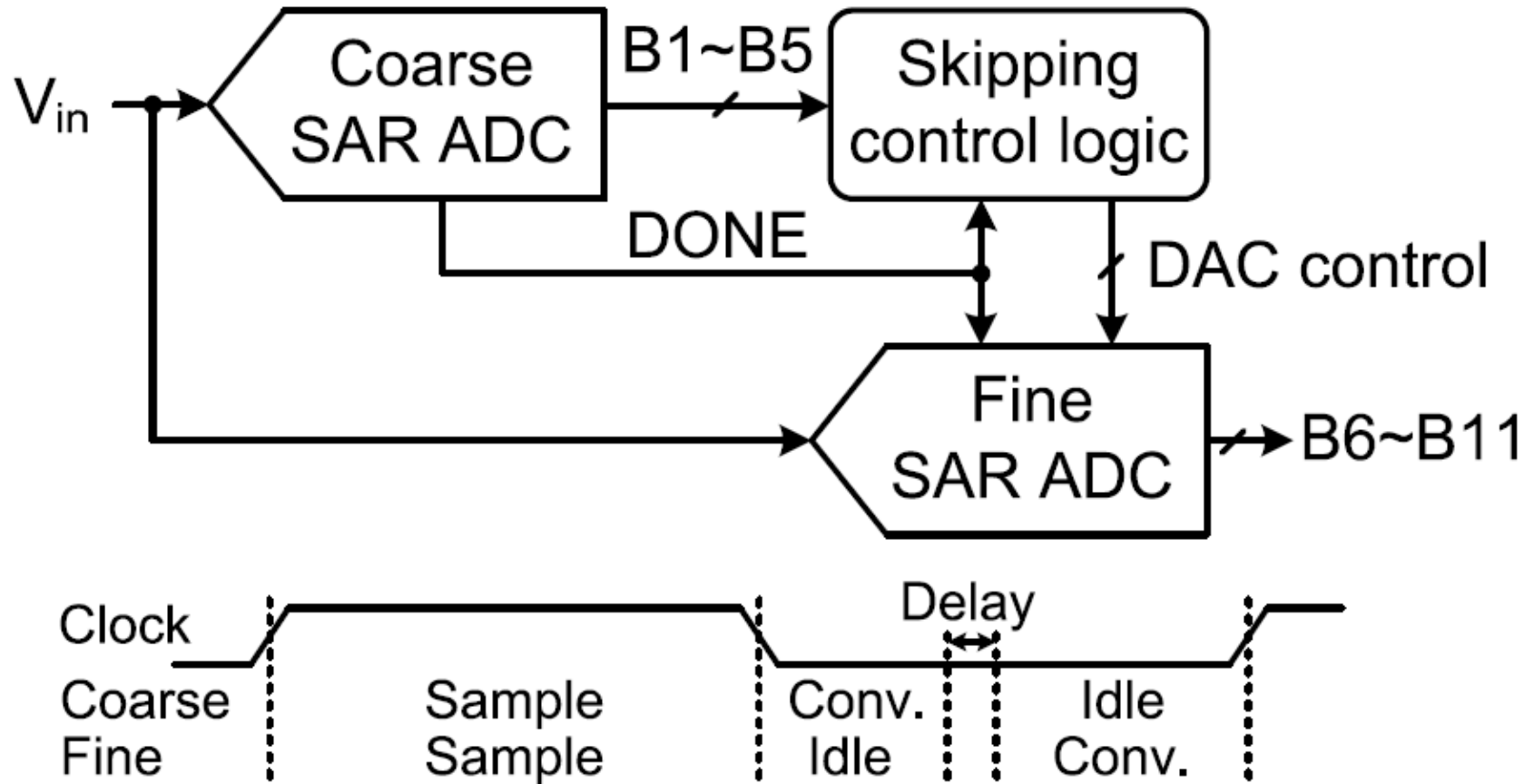


- Standard comparator can be used; offset and noise are reduced

Bottom-plate bidirectional switching



Saving Switching Energy Using Sub-ranging (1)

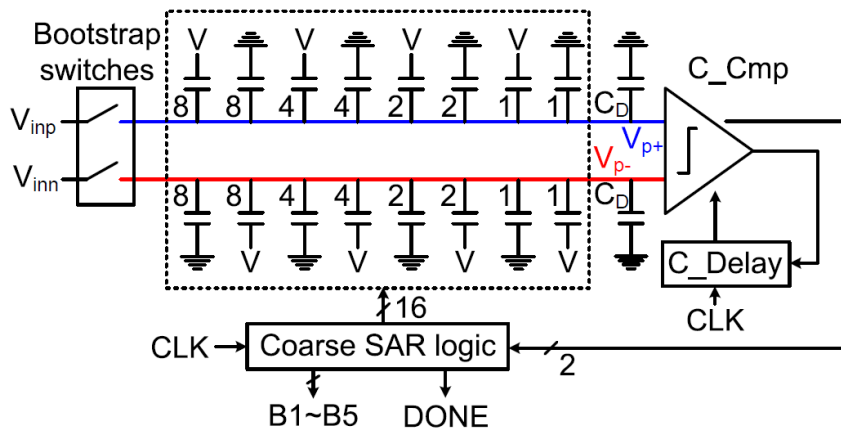


Saving Switching Energy Using Sub-ranging

(2)

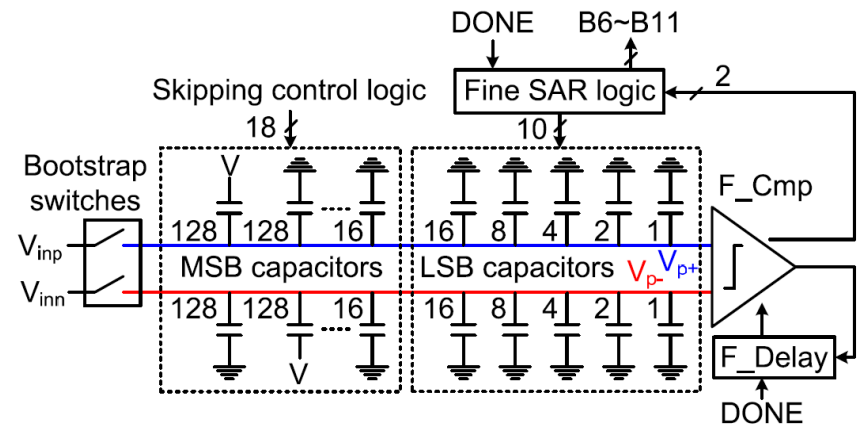
Coarse SAR

Low power comparator



Fine SAR

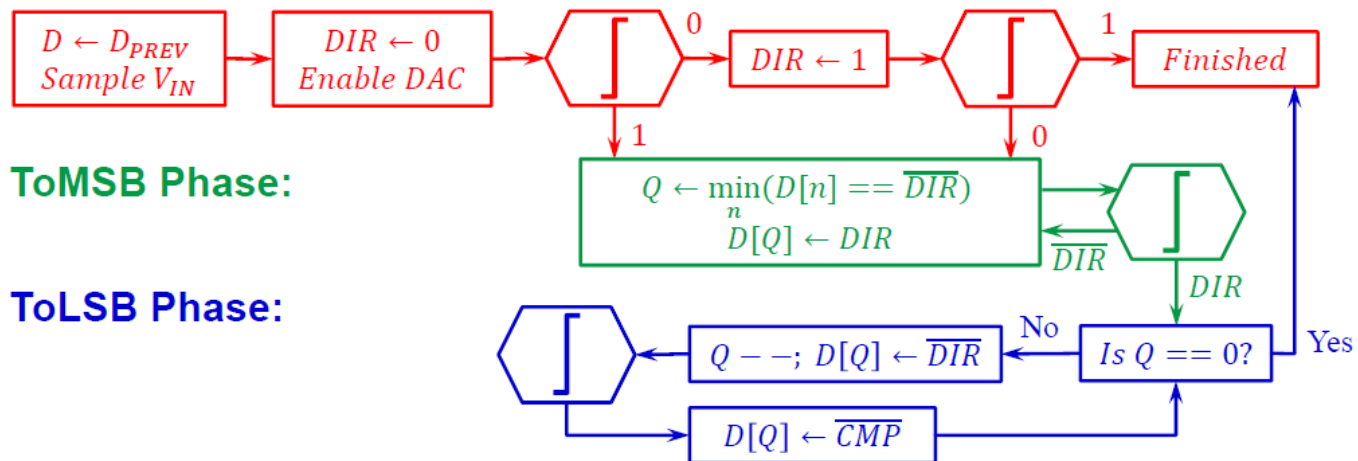
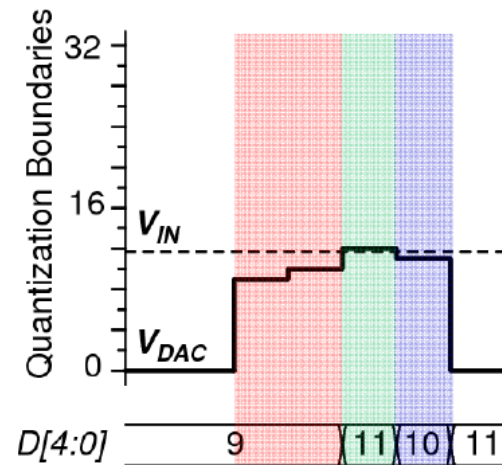
Low noise comparator



LSB-First SAR (1)

$D_{PREV} = 9, D_{OUT} = 11$

Bitcycle	$D[4:0]$	DIR
1	01001	0
2	01001	1
3	01011	1
4	01010	1



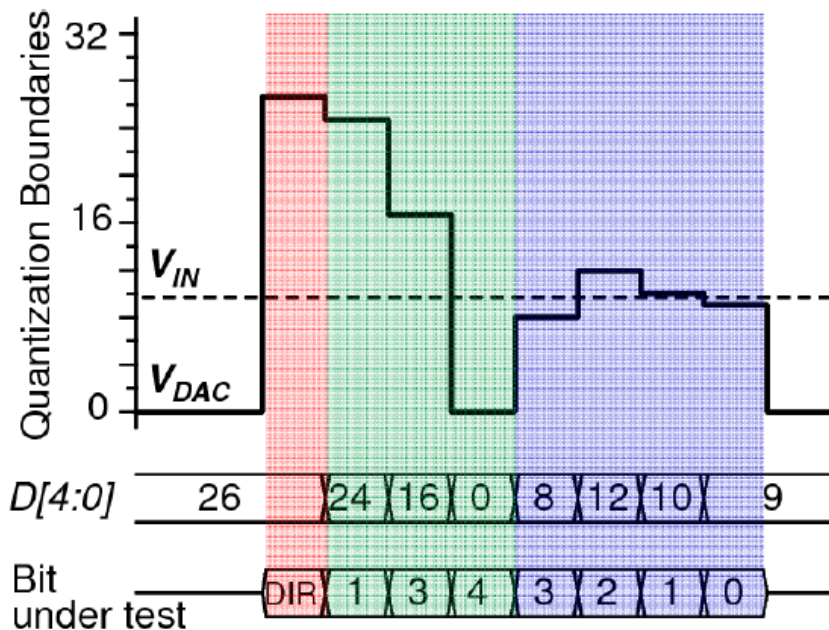
LSB-First SAR (2)

$$D_{PREV} = 26, D_{OUT} = 9$$

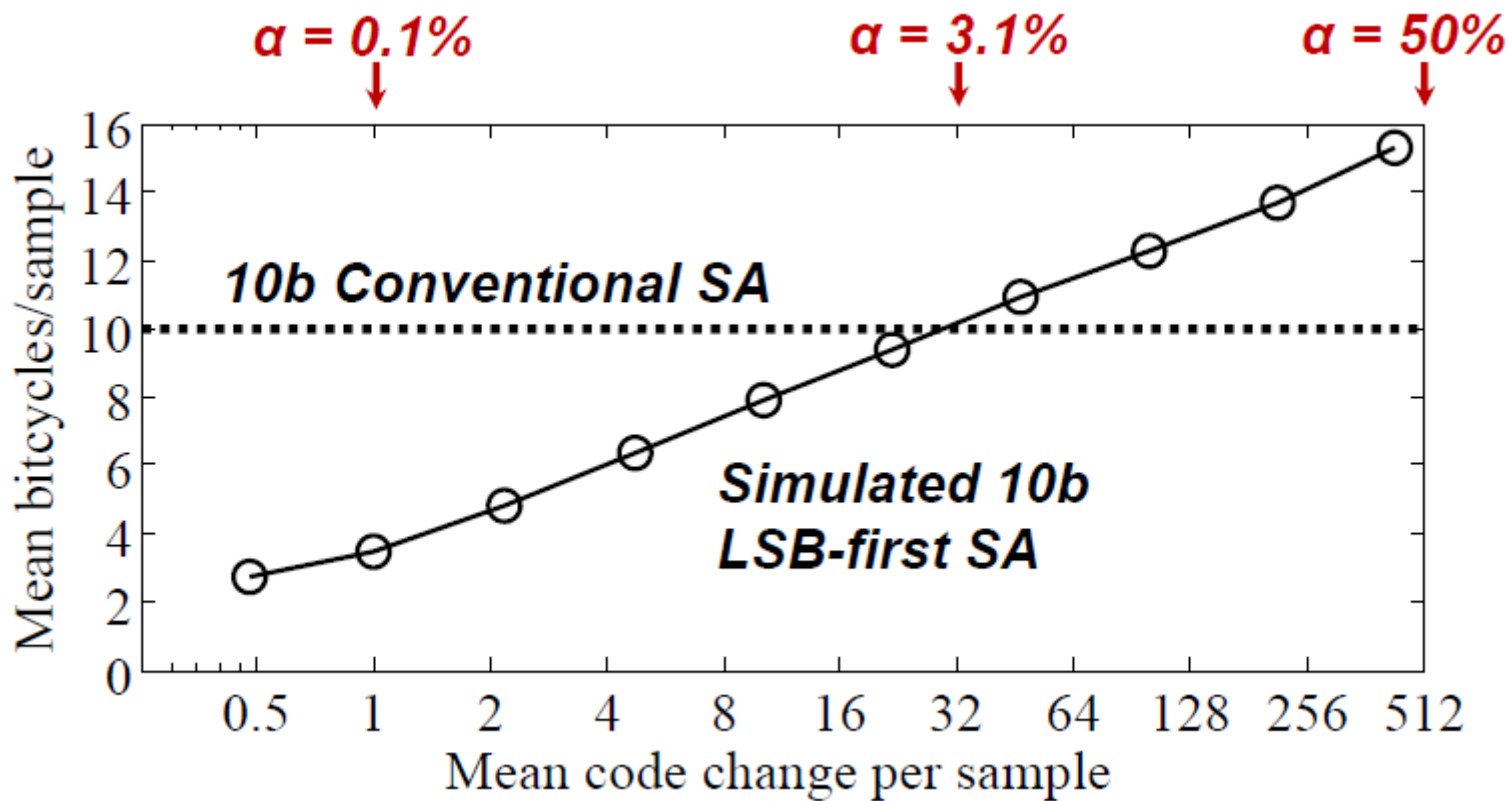
INIT
ToMSB

ToLSB

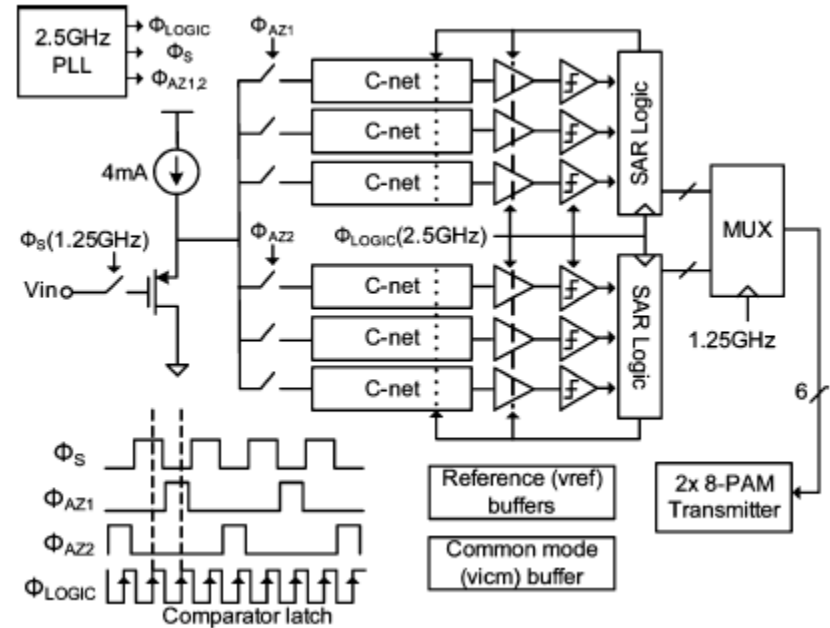
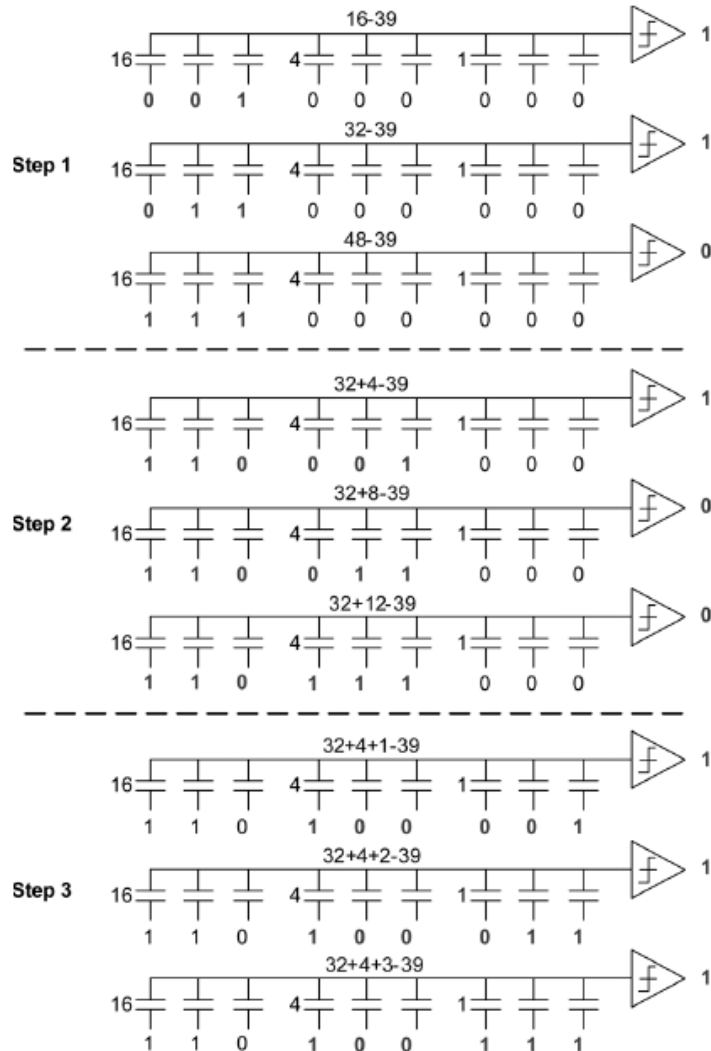
Bitcycle	$D[4:0]$	DIR
1	11010	0
2	11000	0
3	10000	0
4	00000	0
5	01000	0
6	01100	0
7	01010	0
8	01001	0



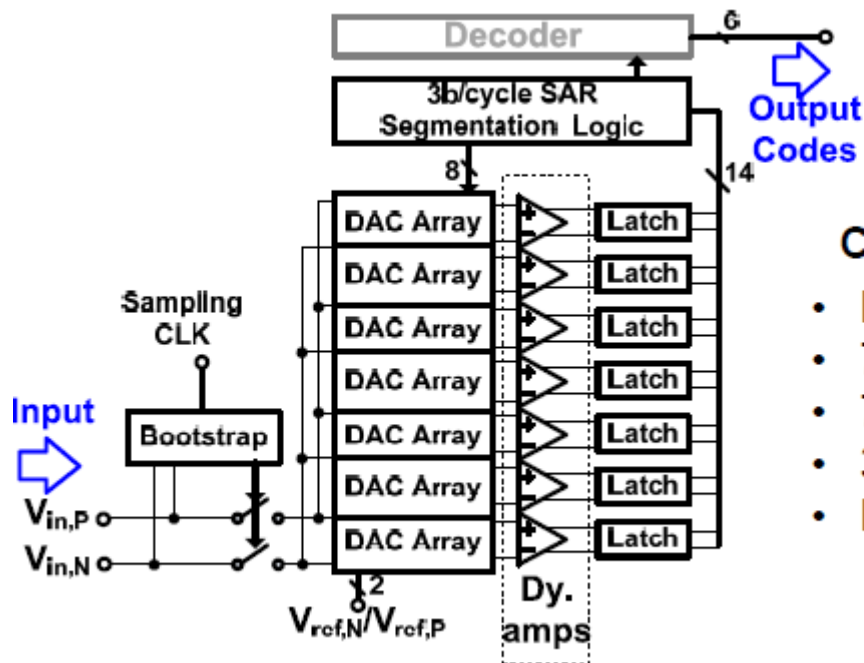
LSB-First SAR (3)



2b/cycle SAR



3b/cycle



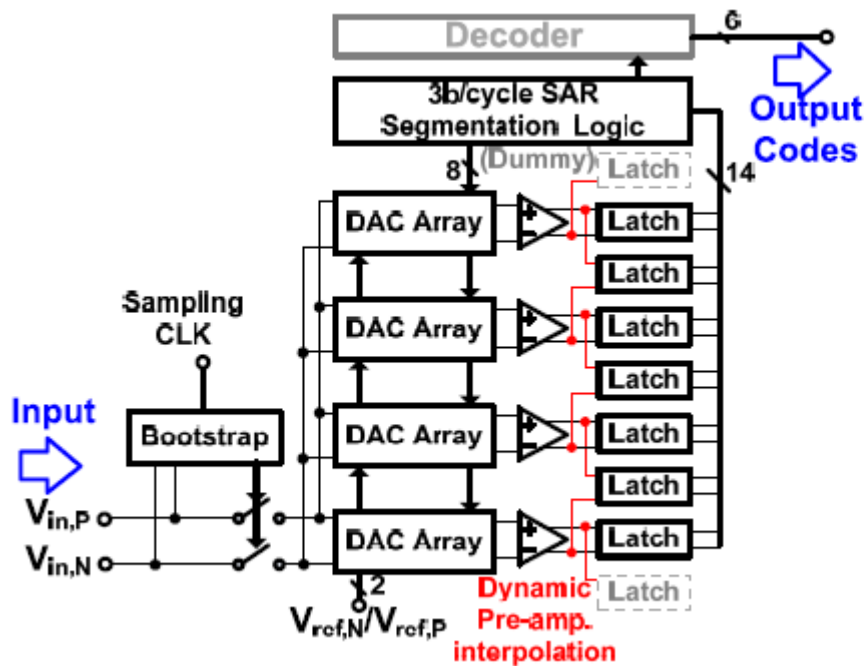
ADC Architecture:

- 3b/cycle SAR

Consist of:

- Bootstrap sampling
- 7 dynamic pre-amps
- 7 DACs & latches
- 3b/cycle logic
- Decoder

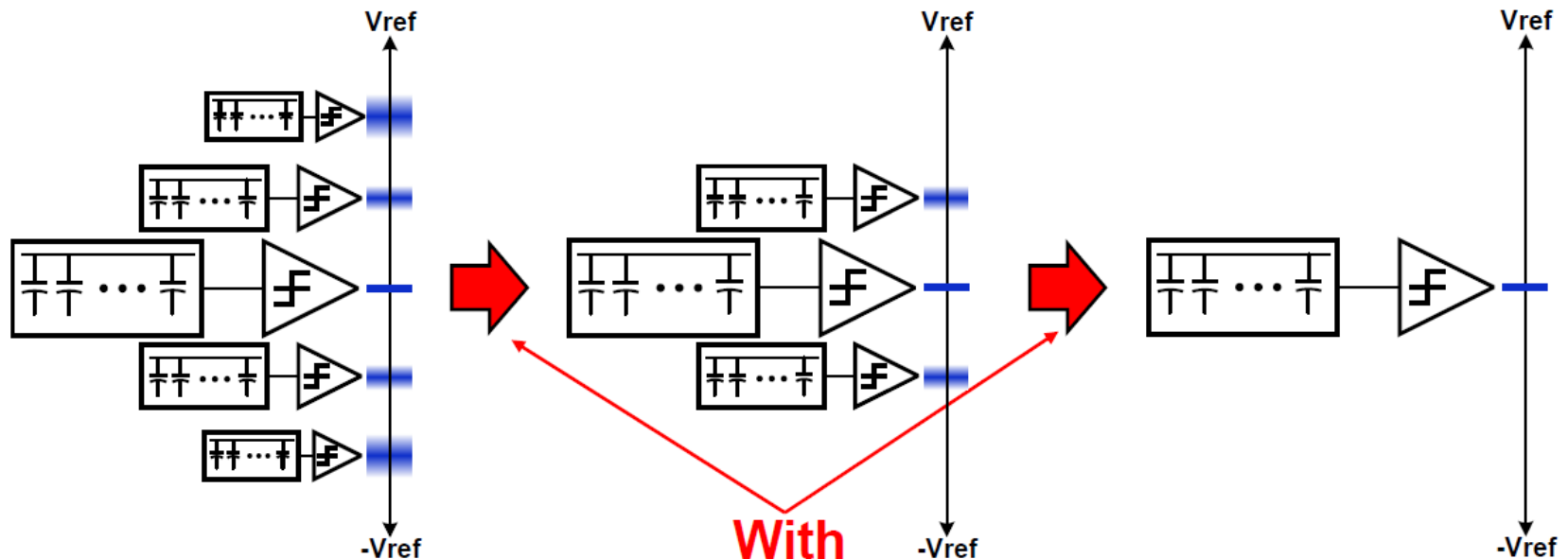
3b/cycle



ADC Architecture:

Hardware saving technique:

Hardware retirement



❑ MSBs

- Low accuracy
- Multi-bits/cycle
 - High speed

❑ LSBs

- High accuracy
- 1b/cycle
 - Low speed

Cap layout

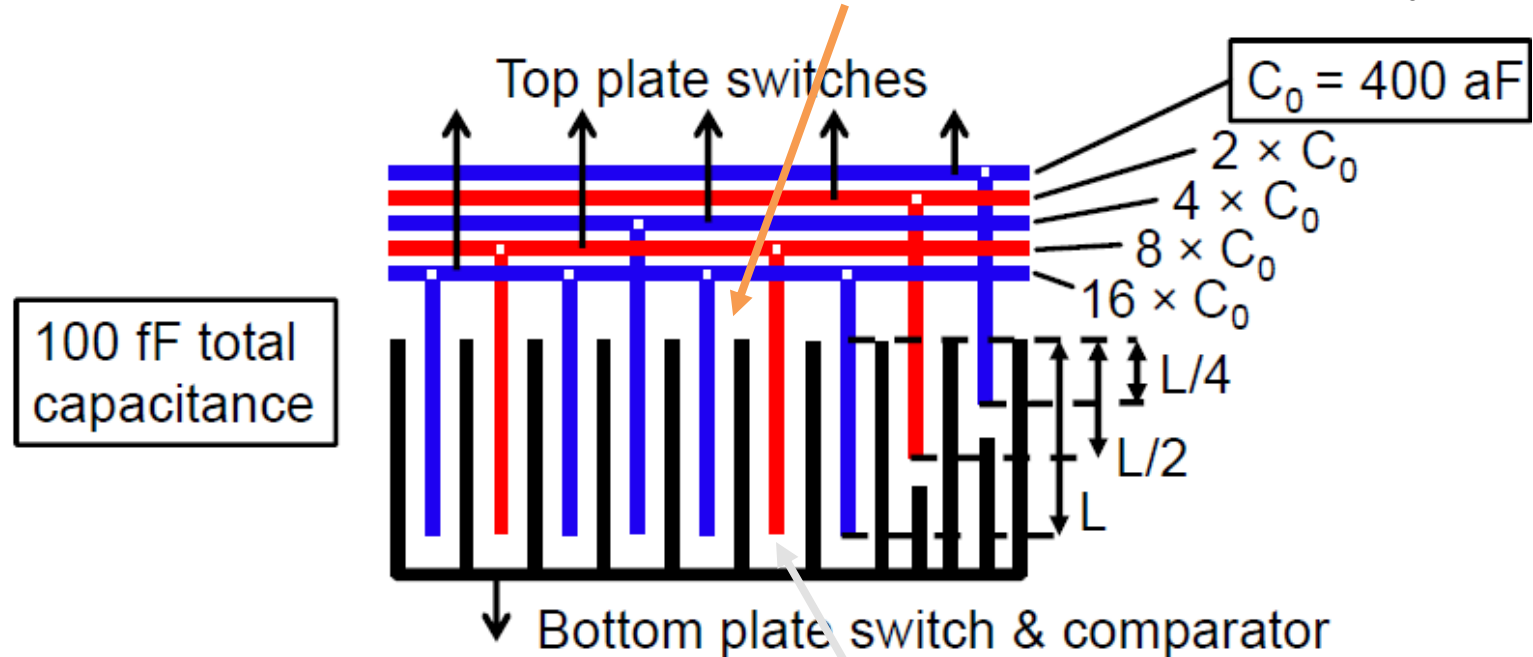
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D	6	5	6	4	6	5	6	3	D
D	6	5	6	4	6	5	6	1	D
D	6	5	6	4	6	5	6	1	D
D	6	5	6	4	6	5	6	3	D
D	6	5	6	4	6	5	6	2	D
D	6	5	6	4	6	5	6	3	D
D	D	D	D	D	D	D	D	D	D

D	D	D	D	D	D	D	D	D	D
D	6	5	6	3	4	6	5	6	D
D	6	5	6	2	4	6	5	6	D
D	6	5	6	3	4	6	5	6	D
D	6	5	6	1	4	6	5	6	D
D	6	5	6	1	4	6	5	6	D
D	6	5	6	3	4	6	5	6	D
D	6	5	6	2	4	6	5	6	D
D	6	5	6	3	4	6	5	6	D
D	D	D	D	D	D	D	D	D	D

Need to consider how we wire the cap out
Need to consider 3D including vias

Low-area layout

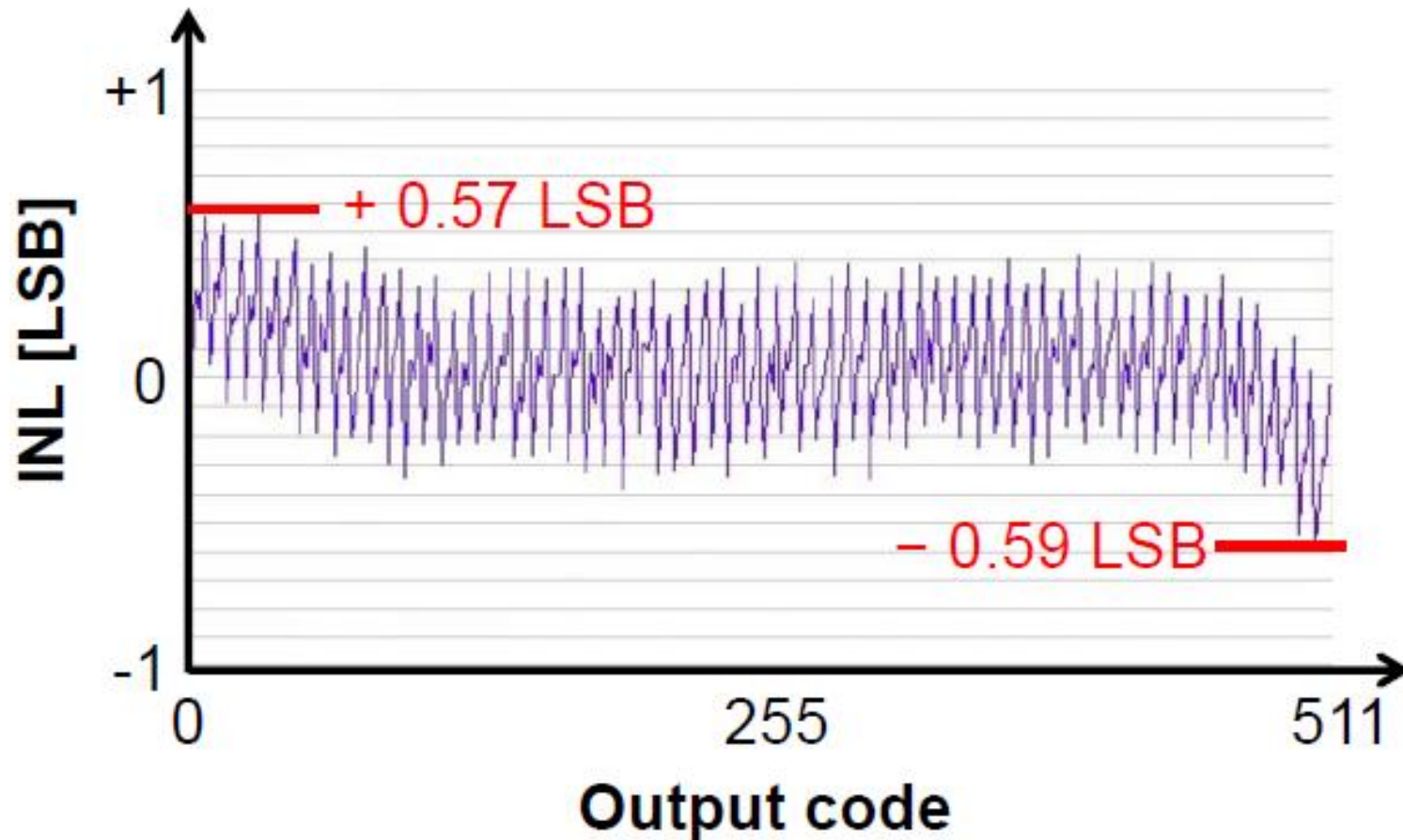
Need shielding on the top, otherwise, there is more cap to $16C_0$



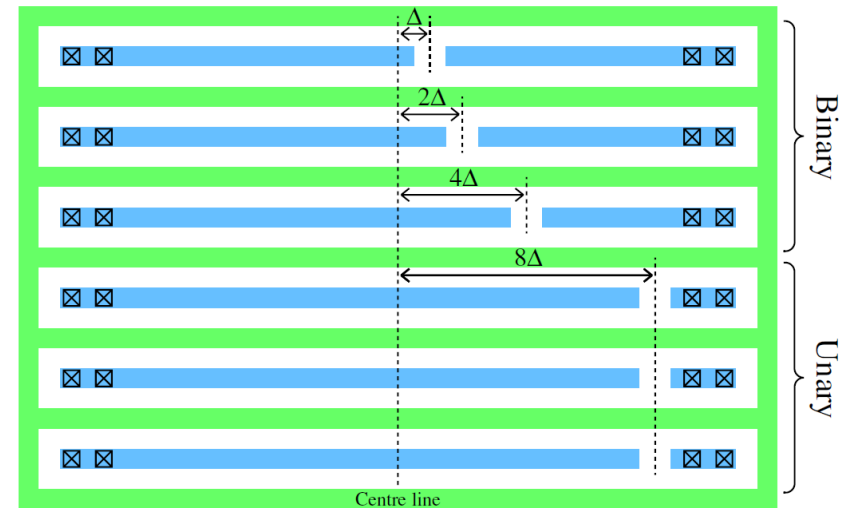
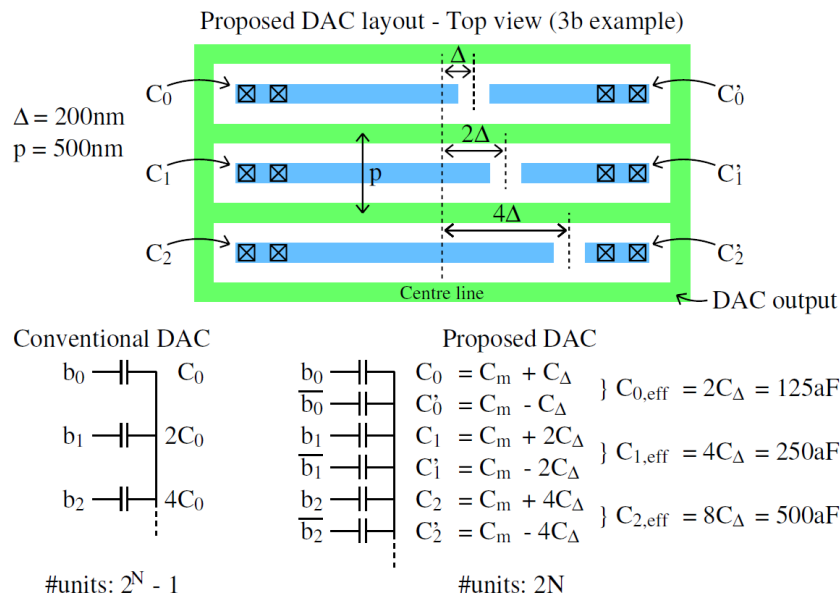
[Dortz, ISSCC, 2014]

Capacitance to the bottom side is not scaled

9-bit ADC measurement data

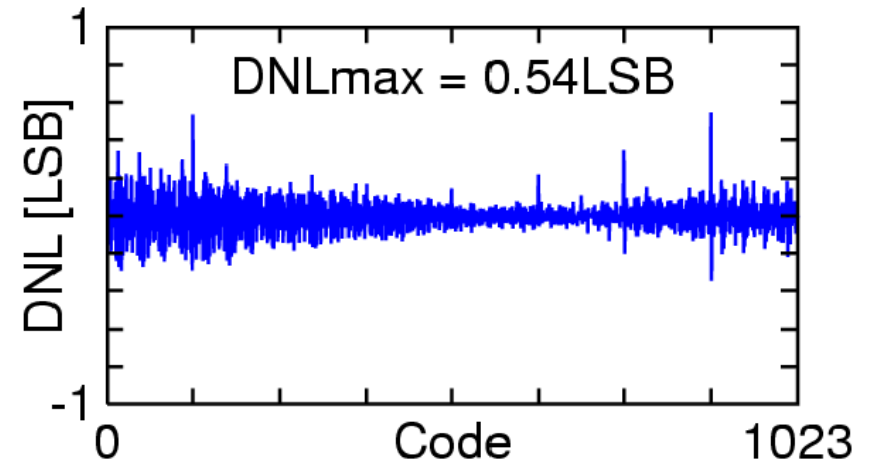
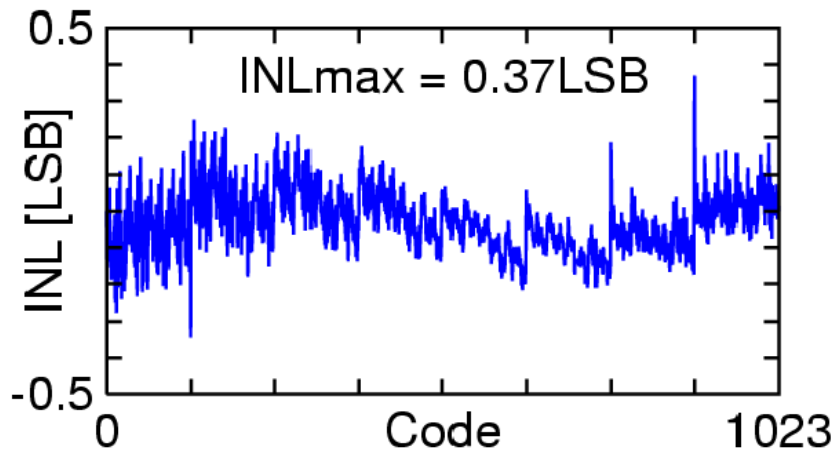


Differential capacitance



[Harpe, CICC, 2018]

10-bit ADC measurement data



Statistics for 10 measured samples			
	INLmax [LSB]	DNLmax [LSB]	ENOB [bit]
Average	0.39	0.55	9.18
Standard deviation	0.03	0.05	0.01

Cap design methodology

- Need to consider the cap from the comparator input to any node
- Need systematic way to ensure matching
- Calibre parasitic extraction cannot be fully relies upon. It uses many approximations to speed up extraction. It is OK for speed/BW estimation, but not matching.
 - To ensure accuracy, we need EM simulation or 3D extraction
 - This can be a good research topic